

Market Concentration and the Productivity Slowdown

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September 5, 2025

Abstract

Since around 2000, U.S. aggregate productivity growth has slowed and product market concentration has risen. To explain these facts, I construct a measure of innovativeness based on patents that is comparable across firms and over time and show that patent quality declined in the 2000s compared to the 1990s, and the decline was larger for market laggards than leaders. I develop an endogenous growth model where the quality of new ideas is heterogeneous across firms to analyze the implications of this finding. A quantitative version of the model is used to infer changes to the structure of the U.S. economy between the 1990s and the 2000s. Changes in the quality of new ideas can account for the bulk of the productivity slowdown and around 40% of the rise in concentration. Strategic changes in firms' R&D investment policies in response to the estimated changes in the quality of new ideas significantly amplify the productivity slowdown.

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1 Introduction

After a boom in the 1990s, U.S. productivity growth declined in the early 2000s and industry leaders began capturing a larger share of sales. I illustrate a new mechanism that can explain both of these trends, along with patterns of increasing profitability, increasing productivity differences, and a declining rate of changes in market leadership in U.S. industries. This mechanism is a decline in the quality of new ideas, particularly for small firms.

I first document that patent quality has fallen since 2000 after a boom in the 1990s. This fact is robust to using market-based measures of patent value or measures of the social value of patents using citations and is broad-based, spanning many sectors of the economy, though it is most pronounced in high tech sectors. An important new finding is that the decline in patent quality is larger for small firms than for large firms: ideas are getting particularly hard for small firms to find.

To understand how leaders (the largest firm in terms of sales in each industry) and laggards (other, smaller firms) respond to the observed changes in patent quality, and the effect of these changes on aggregate growth, I develop and estimate a general equilibrium, quality ladder model of directed innovation along the lines of [Aghion et al. \(2001\)](#). A continuum of industries are populated by two incumbent firms and a competitive fringe producing differentiated goods. Incumbent firms improve their varieties through innovation. Market concentration, measured as the market leader's share of industry sales, is high when the quality difference between the incumbents' products is large.

Unlike [Aghion et al. \(2001\)](#), the model accommodates the possibility that laggard firms have an “advantage of backwardness,” allowing them to improve their variety more drastically than market leaders when they innovate. One parameter governs the extent of laggard firms' advantage of backwardness and another parameter governs the average size of product quality improvements across all firms. Other parameters capture alternative explanations for rising market concentration and slowing productivity growth such as slowing knowledge diffusion, rising market power, increased competition, and declining real interest rates.

To infer the relative importance of different changes to the U.S. economy in explaining the trends of interest, I estimate the model parameters for two steady states

to match data on productivity growth, concentration, aggregate R&D expenditures, the profit share, the rate of turnover in market leadership, and patent quality for the U.S. in two periods, the 1990s and the 2000s. This exercise suggests a significant role for changes in the quality of new ideas, especially a decline in the advantage of backwardness for small firms.

I use the model to explore the channels through which declining idea quality can explain trends in productivity growth and concentration. The estimated changes in idea quality generate a productivity slowdown through two channels. One is direct: even if firms devoted the same resources to research and development, the economy would grow more slowly because average quality improvements are smaller. Second, a strategic effect amplifies the productivity slowdown: because of the relocation of R&D activity towards market leaders, whose innovations tend to be more incremental, the economy grows even more slowly than before. Two decomposition exercises suggest that this latter force accounts for between 15-50% of the productivity slowdown in the model. Quantitatively, the estimated changes in idea quality generate a productivity slowdown in the model of a similar magnitude to the slowdown observed in the U.S.¹

When laggards firms' advantage of backwardness declines, they respond to a lower chance of attaining market leadership by investing less in R&D. Facing a lower probability of being overtaken, leaders invest slightly more. Together, these decisions lead to larger average quality differences between leaders and laggards in steady state, consistent with the data. Quality differences map directly to leaders' market shares and markups, so that sales concentration and markups also rise. Transition dynamics reveal that the decline in idea quality affects concentration slowly: after 25 years changes in idea quality can explain around 40% of the rise in concentration. The dynamics of the transition from high to low idea quality match the relative speeds of the productivity slowdown, which happened quickly, and the rise in concentration, which has been more gradual.

Related Literature This paper contributes a novel mechanism to the large literature linking trends in concentration, productivity growth, and business dynamism

¹The fact that changing innovativeness alone can explain the entire productivity slowdown does not rule out other explanations, since there may be forces working to increase productivity growth that the model does not capture such as population growth, entry, improvements in human capital, and globalization.

using models of endogenous growth. Several papers emphasize the increasing importance of intangible assets and information and communications technology (ICT) as a possible explanation (Aghion et al. (2019), De Ridder (2024), Corhay, Kung, and Schmid (2025), Weiss (2020)), or a more general rise in fixed operating costs for large firms (Ghazi 2021). Rather than emphasizing particular features of ICT, I instead hypothesize that general purpose technologies (GPTs) may affect firm dynamics and market structure while temporarily raising aggregate productivity growth. Past fluctuations in patent quality and productivity growth have been attributed to waves of innovation due to the arrival of new GPTs (Kelly et al. (2021); Kogan et al. (2017)). Given the new evidence presented here on heterogeneity in patent quality across firms and time, I argue that these innovational complementarities appear to be stronger for smaller firms than for market leaders. Non-technological explanations include demographic changes (Hopenhayn, Neira, and Singhania (2018), Jones (2022), Peters and Walsh (2021), Karahan, Pugsley, and Şahin (2024), Engbom (2019), Eggertsson, Mehrotra, and Robbins (2019), Bornstein (2025)) or declining real interest rates (Liu, Mian, and Sufi (2021), Chatterjee and Eyigungor (2023)).

The most closely related explanation is that diffusion of knowledge from leaders to laggards is slowing down (Akcigit and Ates 2021; Akcigit and Ates 2023). The first contribution relative to these papers is the inclusion of a time-varying product differentiation margin, yielding the novel insight that the changes in both business dynamism and productivity growth in recent years could be due to *increased* competition within industries. Second, I develop an indicator of patent quality that maps tightly to the model and target changes in this measure for firms of different sizes in the estimation. Akcigit and Ates (2023) infer a change in knowledge diffusion through changes in entry rates, which may change for many other reasons. Third, aggregate productivity growth is not sensitive to knowledge diffusion in their estimated model, whereas productivity growth is highly sensitive to idea quality in my model, therefore offering a unified explanation for changes in business dynamism and productivity growth. Finally, the two theories have opposite predictions about followers' productivity growth conditional on patenting. I show that this moment in the data is consistent with declining idea quality rather than declining knowledge diffusion.

The model's duopoly formulation also brings together distinct strands of liter-

ature in macroeconomics concerned with (i) slowing growth (ii) changes in market structure and potentially market power and (iii) superstar firms. Strands (ii) and (iii) typically rely on opposing assumptions. According to the literature on rising market power, incumbent firms exercise greater pricing power now than in the past and this is reflected in rising markups and profitability (de Loecker, Eeckhout, and Unger (2020), De Loecker, Eeckhout, and Mongey (2021), Barkai and Benzell (2024)). On the other hand, the literature on superstar firms contends that greater import competition and greater consumer price sensitivity due to better search technology like online retail have *increased* competitive pressures and reduced the market power of incumbent firms, resulting in reallocation to the most productive (superstar) firms (Autor et al. 2020). The model resolves this seeming contrast by demonstrating how markups can rise at the same time as there is reallocation to relatively more productive firms without any changes at all to consumer preferences or the aggregate production function.²

The imperfect substitutes formulation makes it possible to quantitatively compare explanations for increased markups and profits in recent years to the superstar firm hypothesis that greater price sensitivity has sparked reallocation to large, productive firms. Within the model, neither story matches the data as well as a decline in idea quality, though I show that the superstar firm experiment can generate a productivity slowdown alongside rising concentration.

The finding that idea quality has declined since 2000 is consistent with Bloom et al. (2020), who show that despite increasing inputs (expenditures, workers) to R&D, outputs in terms of productivity improvements have declined.³ The main contribution relative to this paper is to show heterogeneity in this fact across the firm size distribution, and analyze the implications of this heterogeneity, which turns out to be important for matching the rise in concentration and the profit share. Several papers have documented that laggard firms are less likely to overtake market leaders

²The model is also consistent with the finding of Kehrig and Vincent (2021) that being a superstar firm is a temporary rather than permanent status. In the model, the relative advantage of high value added firms grows in the 2000s and the average duration of these “shooting star” spells increases, but these firms are eventually displaced by competitors.

³Anzoategui et al. (2019) also identify a decline in R&D productivity using indirect inference in a DSGE model with endogenous productivity growth.

in recent years (Bessen et al. (2020), Pugsley, Sedlacek, and Sterk (2021), Andrews, Criscuolo, and Gal (2016)). Contemporaneous work by Cavenaile, Celik, and Tian (2023) develops an endogenous growth model with step by step innovations and an endogenous number of large firms and finds that declining R&D productivity of small firms can explain a large share of the rise in concentration and the productivity slowdown. This paper provides novel empirical evidence that R&D productivity has indeed declined, particularly for small firms.

Finally, many papers have studied rising concentration and the productivity slowdown (Hall (2015), Syverson (2017)) in isolation from one another. Rising concentration is mainly a within-sector phenomenon (Hsieh and Rossi-Hansberg 2023) that is occurring at the national product market level rather than at the local level.⁴ The finding that concentration is rising is robust to the inclusion of foreign firms (Covarrubias, Gutierrez, and Philippon 2019) or more sophisticated methods of identifying firms' competitors (Pellegrino (2025), using data from Hoberg and Phillips (2016)).⁵

A variety of explanations for rising sales concentration have been proposed, from the introduction of ICT that creates winner-take-all markets and enables the growth of superstar firms (Bessen (2020), Crouzet and Eberly (2018)), to excessive regulations that erect barriers to entry and create unnatural monopolies (Covarrubias, Gutierrez, and Philippon (2019)), to increased mergers and acquisitions activity, possibly due to weak antitrust enforcement (Grullon, Larkin, and Michaely (2019)). This paper complements these hypotheses by contributing a novel mechanism that, according to the quantitative exercise, can potentially explain a significant share of the increase.

⁴In fact Berger, Herkenhoff, and Mongey (2022) and Rossi-Hansberg, Sarte, and Trachter (2020) find evidence that local sales concentration has fallen over this period.

⁵Amiti and Heise (2024) do find that concentration has been stable in manufacturing industries once foreign firms are accounted for but do not expand their analysis to other industries.

2 Empirical Motivation

2.1 Market Concentration and Productivity Growth

Figure 1 plots the average market leader’s share of industry sales in Compustat and the total factor productivity growth rate for the US.⁶ Throughout the paper I focus the analysis on public firms because the model is best suited to address competition among large, incumbent firms and because one of the main moments of interest, the market value of patents, is only available for publicly listed firms. The average market leader’s sales share within narrowly defined 4-digit Standard Industrial Classification (SIC) industries in Compustat has risen from around 40% in the 1990s to over 50% in 2017.⁷ Annual total factor productivity growth averaged about 1.6% between 1994 and 2003, but slowed to about 0.7% on average between 2004 and 2017.

According to the standard [Olley and Pakes \(1996\)](#) decomposition, aggregate total factor productivity growth could be slowing down for two reasons. First, average TFP growth across all firms could be slowing. Second, reallocation to the most productive firms (i.e. the covariance of sales share and productivity) could be slowing. [Baqae and Farhi \(2020\)](#) show that within-firm growth has contributed very little to aggregate TFP growth since the late 1990s while allocative efficiency has risen, lending support to explanations focusing on the incentives of existing firms to increase productivity.

2.2 Trends in Patent Quality

Economists have long relied on patents as an observable proxy for innovativeness ([Griliches 1998](#)). The most commonly used measure of patent quality, counting the number of forward citations a patent receives from future patents, shows substantial heterogeneity in quality in the cross section of patents, with a few patents receiving many citations and most receiving none or just a few ([Akcigit and Kerr 2018](#)).

Recent evidence using alternative measures of patent quality also points to sub-

⁶I focus the analysis on non-financial, non-agricultural firms. See Appendix A.1 for a detailed description of the Compustat sample. Total factor productivity data comes from [Comin et al. \(2025\)](#).

⁷See [Grullon, Larkin, and Michaely \(2019\)](#), [Council of Economic Advisers \(2016\)](#), and [Autor et al. \(2020\)](#) (includes private firms) for overviews of trends in market concentration. More than 75% of U.S. industries have experienced an increase in the Herfindahl-Hirschman index.

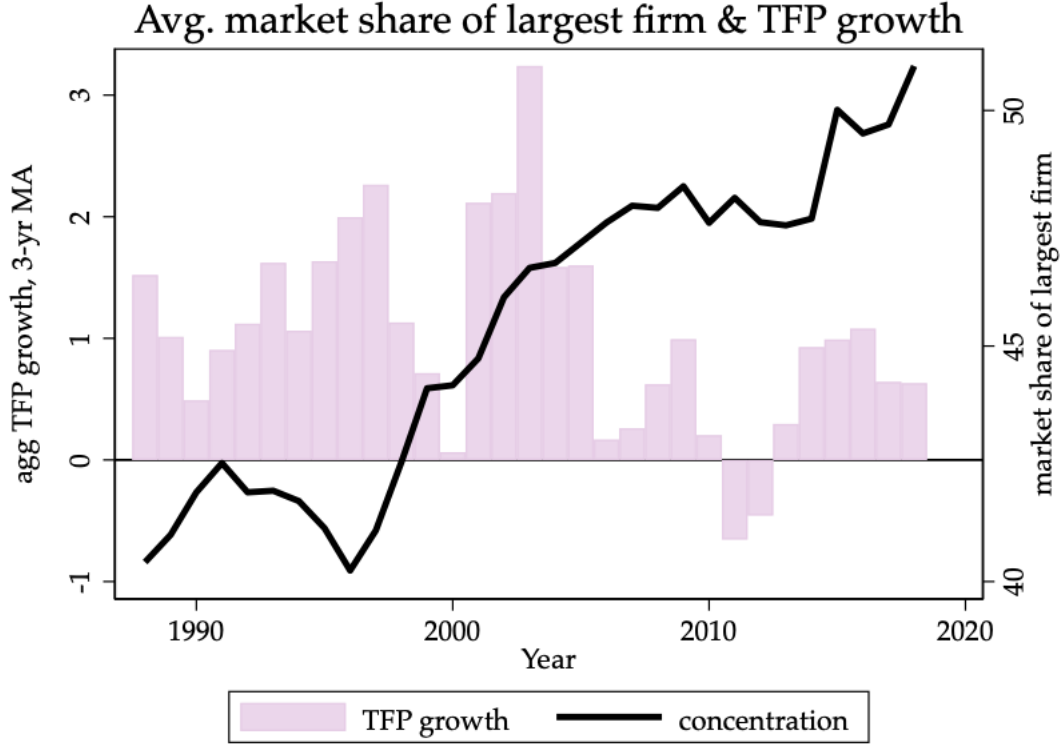


Figure 1: Average market share of largest firm (by sales) in 4-digit SIC industries from Compustat (weighted by industry sales); total factor productivity (TFP) estimates from [Comin et al. \(2025\)](#), three year moving average. See Appendix [A.1](#) for a description of the Compustat sample.

stantial changes in average quality over time. [Kelly et al. \(2018\)](#) create a text-based measure of patent quality, identifying “breakthrough” patents as those patents where the patent’s text differs from the text of past patents but is similar to the text of future patents. This measure has the advantage of covering a longer time series (1860-present) than citation based measures (1940-present). [Kelly et al. \(2018\)](#) find that periods with high average patent quality coincide with the discovery of new general purpose technologies, including the ICT revolution in the 1990s, consistent with [Bresnahan and Trajtenberg \(1995\)](#)’s theory of “innovational complementarities” between general purpose technologies and inventions in other sectors of the economy.⁸

⁸See [Helpman \(1998\)](#), [Aghion, Akcigit, and Howitt \(2014\)](#), and [Brynjolfsson, Rock, and Syverson \(2021\)](#) for reviews of the study of GPTs.

The most recent wave of high patent quality driven by ICT began to subside in the late 1990s according to this measure (Figure A.1 in Appendix A.3).

This paper is the first to explore heterogeneity in the recent decline in patent quality across firms. I use a measure of patent value from Kogan et al. (2017) that estimates the market value of patents issued in the U.S. and assigned to public firms using firms’ excess stock returns in a window around patent approval dates to infer the market value of the patent.⁹ This measure has the advantage of capturing the private value of the patent to the firm, which determines firms’ investment decisions in most models of endogenous growth.¹⁰ Because the estimates use excess returns, the estimates are purged of general movements in the stock market.

In the model in Section 3, firms make innovations that grow their product quality and increase the value of the firm. As firms’ product quality grows, larger and larger improvements in dollar terms are needed to maintain a constant growth rate. I therefore use the dollar value estimates to define *patent quality* as the contribution of a new patent p to the total value of the firm j at time t :¹¹

$$\text{pat. qual.}_{pjt} = \frac{\xi_{pjt}}{M_{j,t-1}}, \quad (1)$$

where ξ_{pjt} is the dollar value estimated by Kogan et al. (2017) and $M_{j,t-1}$ is the one-year lagged market capitalization of firm j . This measure should be stable over time if the economy is on a balanced growth path. From 1994 to 2017 it covers 1,041,136 patents issued to 4,207 different U.S. public firms. A patent may grow the value of a firm because of its effect on product quality, what I will term the “idea quality” of the technology itself. But, this measure also captures the change in the firm’s strategic position within the industry (change in markup, likelihood of imitation, how much the patent disadvantages competitors, etc.) as a result of the patent, which is another

⁹I use the data updated through 2020 from the paper’s webpage:

<https://github.com/KPSS2017/Technological-Innovation-Resource-Allocation-and-Growth-Extended-Data>.

¹⁰Kogan et al. (2017) show that this measure is strongly correlated with forward citation-weighted measures of patent quality at the patent level. Kelly et al. (2021) show that this measure is also correlated with their measure of patent quality based on the novelty of a patent’s text.

¹¹Additional construction and sample details in Appendix A.2. Around 20% of public firms patent in a given year, and 30% ever patent. The patenting firm sample contains manufacturing (80% of firm-year pairs), services (11% of firm-year pairs), and other firms (9%).

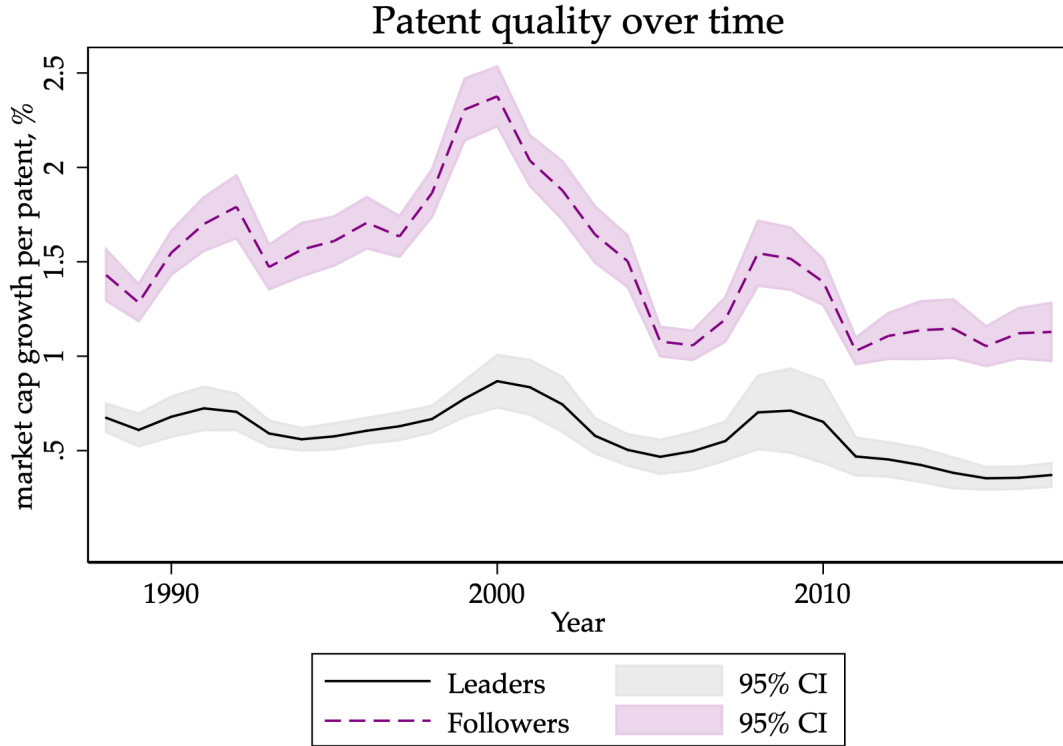


Figure 2: Contribution of average new patent to firm value, using estimated patent values from [Kogan et al. \(2017\)](#). Leader indicates sales leaders in 4-digit SIC industries in a given year and followers are all other public firms in the sample. Three year moving averages. See Appendix [A.1](#) for more details.

incentive for innovation in the model.

Figure 2 plots average patent quality over time, splitting the sample into market leaders (the largest firm by sales in each 4-digit SIC industry) and followers (all other public firms). Figure 2 illustrates two key facts for the subsequent analysis: (1) smaller firms have higher patent quality than market leaders on average; (2) patent quality declined beginning around 2000: followers’ patent quality declined 37.7% and leaders’ declined 28.6%.

Fact (1) is related to the debate on the relative innovativeness of large versus small firms (see [Akcigit and Kerr \(2018\)](#)). For example [Rosen \(1991\)](#) develops a model where large firms optimally focus on innovations that are complementary to their existing products to avoid [Arrow \(1962\)](#)’s replacement effect. Small firms instead fund

the development of disruptive technologies, having more to gain in post-innovation rents from doing so. Typically this debate centers on small startups versus large companies with more than 500 employees (more than 72% of observations in the sample of patenting firms in Compustat have more than 500 employees). I find that even among firms that are large relative to the entire firm size distribution, there are differences in patent quality by size (measured by sales) within industries.

One possible explanation for fact (2) is that general purpose technologies, or at least ICT, have more complementarity with the R&D investments of smaller firms than larger firms (Jovanovic and Rousseau (2005) find that initial public offerings surge during GPT waves, for example). Smaller firms, with greater flexibility and more incentive to invest in riskier, disruptive ideas, may be better positioned to take advantage of the innovation opportunities associated with disruptive technologies. After the GPT has diffused through the economy, opportunities for disruption lessen and laggards' improvements become more incremental. Consistent with this idea, the pattern of boom and bust in patent quality is more pronounced in high tech sectors than in manufacturing, healthcare, or consumer goods, but the trend is present to some extent in all four categories (Figure A.3 in Appendix A.3). Akcigit and Ates (2023) also document a rapid increase in patent self-citation around 2000 for all firms (including private firms), consistent with innovations becoming more incremental and focused around existing technologies. The standard deviation of the patent quality measure declined from 2.5 percentage points (1994-2003) to 1.9 percentage points (2004-2017).

Robustness and Alternative Patent Quality Measures. The sharp decline in patent quality around 2000, especially for laggard firms, is not likely to be driven by the dot-com bubble: the same pattern appears using a similar measure of patent quality based on the contribution of new patents' citations to the stock of past citations (Figure A.2) which do not depend on market valuations.^{12,13} Nor is it driven by

¹²Moreover, as already mentioned, the Kogan et al. (2017) estimates use a firm's individual excess return relative to the market return so is already purged of aggregate market fluctuations.

¹³The only significant change to U.S. patent law in the late 1990s was the American Inventor's Protection Act of 1999 to publish patent applications after 18 months. Previously, only approved patents were published. This change might deter applications where a patent was unlikely to be

ideas being increasingly embodied in multiple patents: the annual increase in market capitalization attributable to patenting measured by summing all patents within the year has also declined (Appendix Table A.3). There has also been a decline in the annual dollar value of patents generated per dollar spent on research and development (Appendix Table A.3). The dollar value estimates of patent quality (Kogan et al. 2017), without any deflator, also show a declining trend (Appendix Figure A.5). The patent quality measure in Figure 2 is used as the measure of patent quality in the model in the next section since it is most tightly linked to innovation incentives in the model.

2.3 Declining Dynamism and Leadership Turnover

In the model, innovations drive growth in market share at the expense of the firm’s competitors. I compute the fraction of industries in Compustat with a new largest firm in terms of sales each year to measure the frequency with which smaller firms overtake the largest firm. This fraction has fallen from 15% per year in the late 1990s to 9%. See Bessen et al. (2020) for a detailed empirical analysis of this phenomenon in the U.S. and Andrews, Criscuolo, and Gal (2016) for a cross-country analysis.

Firm-level productivity data also shows that the “advantage of backwardness” has fallen relative to the 1990s. Andrews, Criscuolo, and Gal (2016) show that in a regression of firm-level productivity growth on a variety of explanatory variables, the coefficient on the lagged productivity gap to the most productive competitor has decreased over the 2000s, suggesting that distance to the productivity frontier is becoming a less important predictor of future productivity growth. Decker et al. (2016) also find that the right skewness of the firm-level productivity growth distribution in the U.S. has declined over this period. The dispersion of revenue-based total factor productivity growth conditional on patenting has also declined in recent years (see Appendix A.6.1 for further discussion).

approved for fear that the idea would be published but not patented. In that case one would expect the patent approval rate to rise, but the approval rate declined from 70% in 1996 to 40% in 2005 (Carley, Hegde, and Marco 2015). Moreover, firms given the option to opt out of this pre-grant disclosure chose to do so less than 10% of the time (Graham and Hegde 2015).

3 Model

To study the growth and market structure effects of declining patent quality, I develop a model along the lines of [Aghion et al. \(2001\)](#) but building on models with heterogeneous patent quality rather than step by step innovations ([Akcigit, Ates, and Impullitti \(2025\)](#), [Acemoglu et al. \(2018\)](#), [Akcigit and Kerr \(2018\)](#)) to allow for heterogeneity in the decline between the largest firms (leaders) and the rest (followers). Relative to [Aghion et al. \(2001\)](#), I also introduce a competitive fringe of firms in each sector that constrains the pricing behavior of the incumbents in order to match levels of concentration in the data. Markups are endogenous and each sector's level of sales concentration varies over time as the result of innovation. The model is of a closed economy in continuous time. There are three types agents: a representative household, a representative final good firm, and firms producing intermediate goods.

3.1 Households

A representative household consumes, saves, and supplies labor to maximize:

$$U_t = \int_t^\infty \exp(-\rho(s-t)) \frac{C_s^{1-\psi}}{1-\psi} ds,$$

subject to:

$$\begin{aligned} r_t A_t + W_t L &= P_t C_t + \dot{A}_t, \\ A_t &= \int_0^1 \sum_{i=1}^2 (V_{ijt} + V_{ijt}^e) dj, \end{aligned}$$

where V_{ijt} is the value of an incumbent intermediate good firm i in sector j at time t and V_{ijt}^e is the value of an entrant that can displace firm i in sector j at time t . These value functions are explained in greater detail in section 3.3. r_t is the rate of return on the portfolio of firms. On a balanced growth path with constant growth rate of output g this yields the standard Euler equation $r = g\psi + \rho$.

3.2 Final Good Producers

The competitive final goods sector combines intermediate goods and labor to create the final output good which is used in consumption, research, and intermediate good production. The final good firm operates a constant return to scale technology:

$$Y_t = \frac{1}{1-\beta} \left(\int_0^1 K_{jt}^{1-\beta} dj \right) L^\beta, \quad (2)$$

where K_{jt} is a composite of three products produced by sector j described below. β determines the elasticity of substitution across sectors ($\frac{1}{\beta}$) and the labor share.¹⁴ The final good firm hires sector composite goods K_{jt} for $j \in [0, 1]$ and labor L by solving:

$$\max_{K_{jt}, L} P_t \frac{1}{1-\beta} \left(\int_0^1 K_{jt}^{1-\beta} dj \right) L^\beta - \int_0^1 P_{jt} K_{jt} dj - W_t L.$$

The first order condition for sector j 's composite good given sector j 's composite price index P_{jt} yields the demand for sector j 's good:

$$K_{jt} = \left(\frac{P_{jt}}{P_t} \right)^{-\frac{1}{\beta}} L,$$

and the real wage is equal to the marginal product of labor:

$$\beta \frac{Y_t}{L} = \frac{W_t}{P_t}.$$

To derive the demand curve for each intermediate good producer i in sector j we need to define the sector composite goods K_{jt} explicitly:

$$K_{jt} = \left((q_{1jt} k_{1jt})^{\frac{\epsilon-1}{\epsilon}} + (q_{2jt} k_{2jt})^{\frac{\epsilon-1}{\epsilon}} + (q_{fjt} k_{fjt})^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}}, \quad (3)$$

where q_{ijt} is the quality of firm i 's product at time t (equivalently firm i 's productivity) and k_{ijt} is the output of firm i purchased by the final good producer.¹⁵ The elasticity of substitution between product varieties in the same sector is ϵ .

The first order condition for the final good firm's problem yields the following demand curve for firm i in sector j 's output:

$$k_{ijt} = q_{ijt}^{\epsilon-1} \left(\frac{p_{ijt}}{P_{jt}} \right)^{-\epsilon} \left(\frac{P_{jt}}{P_t} \right)^{-\frac{1}{\beta}} L. \quad (4)$$

That is, demand is increasing in the firm's quality, decreasing in its price relative to the sector j price index, and decreasing in the sector's relative price. Throughout, the price of the final good P_t is normalized to one and L is set to 1.

¹⁴Section 4.2 discusses how a standard calibration of the labor share implies a cross-sector elasticity of substitution that is consistent with estimates of this parameter in the data.

¹⁵I use quality and productivity interchangeably because final output is homogeneous of degree one in either the qualities or the quantities of the intermediate goods firms' products and labor.

3.3 Intermediate Goods Producers

Each intermediate good sector features competition between two large firms with differentiated products and access to an R&D technology, plus a competitive fringe. The large firms are occasionally hit with exit shocks that cause them to be replaced by a potential entrant.

3.3.1 Production and Price Setting

Production. Intermediate goods producers purchase final goods to transform them into differentiated intermediate goods. Each unit of intermediate output requires η units of the final good to produce. There are no other inputs to intermediate good production.

Competitive fringe. Each industry has a competitive fringe of firms that can imperfectly imitate the incumbent firm with the lower quality product (the follower). The fringe's quality $q_{fjt} = \zeta q_{ijt}$, where $i \in \{1, 2\}$ is the identity of the incumbent firm in industry j with lower quality at time t . Assuming the presence of a competitive fringe is not necessary to solve the model, but makes it possible to match the average level of sales concentration across sectors in the data and generates plausible predictions for profit shares as a function of market shares (see Appendix A.5.)¹⁶

Price setting. Firms set prices a la Bertrand at each instant t . Since the fringe firms all produce the same variety with the same quality and marginal cost, these firms are price takers and set $p_{fjt} = \eta$. Total production by the fringe k_{fjt} is pinned down by demand from the final good producer.

Omitting the subscript t , the static problem of large firm i in sector j is:

$$\max_{p_{ij}} p_{ij} k_{ij} - \eta k_{ij},$$

subject to the demand:

$$k_{ij} = q_{ij}^{\epsilon-1} \left(\frac{p_{ij}}{P_j} \right)^{-\epsilon} (P_j)^{-\frac{1}{\beta}} L,$$

¹⁶I solve a version of the model without the competitive fringe in Appendix B.6 and replicate the main exercises in this setting. The main results are qualitatively unchanged.

where

$$P_j = \left(q_{1j}^{\epsilon-1} p_{1j}^{1-\epsilon} + q_{2j}^{\epsilon-1} p_{2j}^{1-\epsilon} + q_{fj}^{\epsilon-1} \eta^{1-\epsilon} \right)^{\frac{1}{1-\epsilon}}$$

is sector j 's price index.

Let $s_{ij} = \frac{p_{ij} k_{ij}}{p_{1j} k_{1j} + p_{2j} k_{2j} + \eta k_{fj}}$ denote firm i 's market share in sector j . Then the optimal pricing policy for the large firms is:

$$p_{ij} = \frac{\epsilon - (\epsilon - \frac{1}{\beta}) s_{ij}}{\epsilon - (\epsilon - \frac{1}{\beta}) s_{ij} - 1} \eta. \quad (5)$$

The optimal price is the standard two-layered constant elasticity of demand (nested CES) solution: a variable markup rising in market share.

3.3.2 Innovation

The two large firms in each industry have access to a research and development technology that allows them to choose an amount of research spending R_{ijt} of the final good to maximize the discounted sum of expected future profits. The decision to model R&D as a process of own-product quality improvement by incumbents is consistent with the evidence in [Garcia-Macia, Hsieh, and Klenow \(2019\)](#) that: (i) incumbents are responsible for most employment growth in the U.S., and this share has increased in recent years; (ii) growth mainly occurs through quality improvements rather than new varieties; (iii) creative destruction by entrants and incumbents over other firms' varieties accounted for less than 25% of employment growth from 2003-2013, consistent with earlier evidence from [Bartelsman and Doms \(2000\)](#). This choice also enables mapping to the patent quality data which only covers relatively large, public firms. The model still features productivity growth from firms behind the quality frontier because varieties are differentiated and quality improvements to laggards' varieties therefore increase aggregate productivity.

Innovations arrive randomly at Poisson rate x_{ijt} which depends on research spending according to the function:

$$x_{ijt} = \left(\frac{\gamma R_{ijt}}{\alpha} \right)^{\frac{1}{\gamma}} q_{ijt}^{\frac{1-\frac{1}{\beta}}{\gamma}}.$$

Since $\beta < 1$, with higher quality more R&D is needed to achieve the same arrival rate x . γ and α are R&D technology parameters.

Innovations improve the quality of the firm's own variety.¹⁷ Conditional on innovating, the size of the quality improvement is random. Formally,

$$q_{ij(t+dt)} = \lambda^{n_{ijt}} q_{ijt},$$

where $\lambda > 1$ is some minimum quality improvement and $n_{ijt} \in \mathbb{N}$ is a random variable. Initial qualities of all firms at $t = 0$ are normalized to 1. Let $N_{ijt} = \int_0^t n_{ijs} ds$ denote the total number of λ step improvements over a product line i since the beginning of time. The *technology gap* m_{ijt} from firm 1 in sector j 's perspective at time t is defined as:

$$\frac{q_{1jt}}{q_{2jt}} = \frac{\lambda^{N_{1jt}}}{\lambda^{N_{2jt}}} \equiv \lambda^{m_{1jt}}.$$

Given λ , m_{ijt} parameterizes the relative qualities of the two large firms in sector j from firm $i \in \{1, 2\}$'s perspective, representing the number of λ steps ahead of or behind its competitor firm i is. m_{ijt} turns out to be the only payoff relevant state variable for the firms. For tractability I will impose a maximal technology gap $\bar{m}$, but in calibrating the model the parameters are such that this maximal gap rarely occurs in steady state. The only knowledge spillover between the large firms in the model occurs when a firm at the maximal gap innovates. In that case, both the innovating firm and its competitor's quality increase by the factor λ , keeping the technology gap unchanged but raising the absolute quality of the sector composite good.

The distribution of possible quality improvements depends on the firm's current technology gap, consistent with the evidence in Section 2 that patent quality varies between market leaders and laggards. Because of the subsequent result that firm strategies depend on technology gaps, it is useful to formulate these distributions as though firms draw a new position in technology gap space $n \in \{-\bar{m}, \dots, \bar{m}\}$ when they innovate, rather than an absolute number of λ steps. Given n and the current technology gap m the number of steps can then be derived as $n - m$.

As in [Akcigit, Ates, and Impullitti \(2025\)](#), I assume there exists a fixed distribution $\mathbb{F}(n) \equiv c_0(n + \bar{m})^{-\phi}$ for all $n \in \{-\bar{m} + 1, \dots, \bar{m}\}$ that applies to firms that are

¹⁷See [Griliches \(2001\)](#) for a survey of the relationship between R&D and productivity at the firm level and [Zachariadis \(2003\)](#) for a leading empirical test. [Luttmer \(2007\)](#) provides an additional rationale for this assumption: entrants are usually small and enter far from the productivity frontier, implying that imitation of others is difficult.

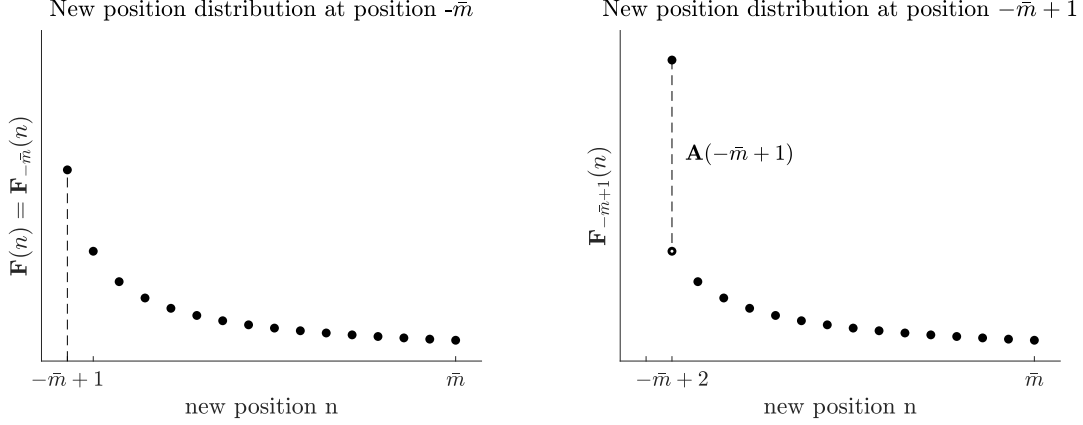


Figure 3: Examples of new position distributions for positions $-\bar{m}$ and $-\bar{m} + 1$.

the furthest possible distance behind their competitor, that is, at position $-\bar{m}$, and describes the probability that they move to each position in technology gap space. An example is shown in the left panel of Figure 3. The shape parameter ϕ is critical in the model and determines the speed of catchup by increasing or decreasing the relative probability of larger innovations. A higher ϕ means a lower probability of these “radical” improvements.^{18,19} c_0 is simply a shifter to ensure $\sum_n \mathbb{F}(n) = 1$.

Given this fixed distribution for the most laggard firm, firms at position $m > -\bar{m}$ draw their new position from the distribution:

$$\mathbb{F}_m(n) = \begin{cases} \mathbb{F}(m+1) + \mathbb{A}(m) & \text{for } n = m+1 \\ \mathbb{F}(s) & \text{for } n \in \{m+2, \dots, \bar{m}\} \end{cases},$$

where $\mathbb{A}(m) \equiv \sum_{-\bar{m}+1}^m \mathbb{F}(n)$. This distribution is shown in the right panel of Figure 3 for a firm at gap $-\bar{m} + 1$. All the mass of the fixed distribution on positions lower than the current position m is put on one-step ahead improvements. This formulation captures the feature that laggard firms make larger improvements than leaders on average. A change in the shape parameter ϕ affects the quality improvement distribution facing all firms, but impacts firms with the lowest relative quality the most

¹⁸As noted by [Akcigit, Ates, and Impullitti \(2025\)](#), this formulation converges to the step-by-step model as $\phi \rightarrow \infty$.

¹⁹The use of “radical innovation” in this paper to describe a relatively large quality improvement differs from some other papers in the literature such as [Acemoglu and Cao \(2015\)](#) who use “radical innovation” to refer to an entrant replacing an incumbent.

and has a decaying impact on firms with higher and higher relative quality.

3.3.3 Entry and Exit

The incumbent firms face a constant exit risk δ_e . If hit with this shock, a firm is replaced by an entrant that takes over the product line with the same quality level as the incumbent. This shock captures reasons why incumbents may be displaced by entrants that are not related to the incumbent firms' innovations, such as knowledge diffusion or imitation (Akçigit and Ates 2023). The mapping to the knowledge diffusion shocks in Akçigit and Ates (2023), which hit only the technology leader and benefit the follower, changes when goods are differentiated. In the present model, both large firms are subject to knowledge diffusion shocks since their products differ and potential entrants can imitate either firm. Akçigit and Ates (2023) document rising self-citation and shorter claim length in patents among all firms, and show that the top 1% of patenting firms have increased their share of patents since 2000 as reasons knowledge diffusion may have slowed, but this does not suggest these trends differ among, say, the two largest firms in an industry.

3.3.4 Intermediate Goods Firms Value Functions

Turning to the firm value functions, I show that the technology gap $m \in \{-\bar{m}, \dots, \bar{m}\}$ is sufficient to describe the firms' pricing and innovation strategies, and that firm values scale in a particular function of their current product quality q_{ijt} .

The proof that pricing decisions and market shares depend only on m_{ijt} (and not on the level of quality q_{ijt}) is in Appendix B.1. The flow profits of the large firms, denoting the optimal price at technology gap m as p_m , are:

$$\begin{aligned} \pi(m, q_{ijt}) &= (p_m - \eta)k_{ijt} \\ &= \begin{cases} q_{ijt}^{\frac{1}{\beta}-1} (p_m - \eta) p_m^{-\epsilon} (p_m^{1-\epsilon} + (\lambda^{-m})^{\epsilon-1} p_{-m}^{1-\epsilon} + (\zeta)^{\epsilon-1} \eta^{1-\epsilon})^{\frac{\epsilon-\frac{1}{\beta}}{1-\epsilon}} & \text{for } m \in \{-\bar{m}, \dots, 0\} \\ q_{ijt}^{\frac{1}{\beta}-1} (p_m - \eta) p_m^{-\epsilon} (p_m^{1-\epsilon} + (\lambda^{-m})^{\epsilon-1} p_{-m}^{1-\epsilon} + (\zeta \lambda^{-m})^{\epsilon-1} \eta^{1-\epsilon})^{\frac{\epsilon-\frac{1}{\beta}}{1-\epsilon}} & \text{for } m \in \{1, \dots, \bar{m}\} \end{cases} \end{aligned}$$

where the second line uses equation 4 and the definition of the sector price index.

For the dynamic problem, I use a guess and verify method to verify that firms' strategies depend only on m and that firm values scale in some function of q_{ijt} . Given

an interest rate r_t , the value function of a firm with gap m to its competitor and quality level q_{ijt} can be written:

$$\begin{aligned}
r_t V_{mt}(q_{ijt}) - \dot{V}_{mt}(q_{ijt}) = & \max_{x_{mt}} \left\{ \pi(m, q_{ijt}) - \alpha \frac{(x_{mt})^\gamma}{\gamma} q_{ijt}^{\frac{1}{\beta}-1} \right. \\
& + x_{mt} \sum_{n_t=m+1}^{\bar{m}} \mathbb{F}_m(n_t) [V_{nt}(\lambda^{n_t-m} q_{ijt}) - V_{mt}(q_{ijt})] \\
& + x_{(-m)t} \sum_{n_t=-m+1}^{\bar{m}} \mathbb{F}_{-m}(n_t) [V_{(-n)t}(q_{ijt}) - V_{mt}(q_{ijt})] \\
& \left. + \delta_e (0 - V_{mt}(q_{ijt})) \right\}. \tag{6}
\end{aligned}$$

The firm chooses the arrival rate of innovations x_{mt} . The first line shows the flow profits and the R&D cost R_{ijt} given the choice of x_{mt} . The second line gives the probability that the firm innovates and sums over the possible states the firm could move to using the new position distribution and the firm's new value function with a higher level of product quality *and* a higher *relative* quality compared to its industry rival. The third line denotes the chance that the firm's rival innovates and the change in the firm's value because its relative quality falls. The final line shows the chance that the large firm is displaced by a new entrant. The slightly altered equations for firms at the minimum and maximum gaps due to knowledge spillovers are given in Appendix B.2.

A guess and verify approach verifies that $V_{mt}(q_{ijt}) = v_{mt} q_{ijt}^{\frac{1}{\beta}-1}$. Thus one can focus on a Markov perfect equilibrium where firms' strategies depend only on the payoff-relevant state variable m which characterizes the technology gap between the two large firms.

The firm's optimal innovation rate x_{mt} is the solution to the first order condition of equation (6), which gives:

$$x_{mt} = \begin{cases} \left(\frac{\sum_{n_t=m+1}^{\bar{m}} \mathbb{F}_m(n_t) [(\lambda^{n_t-m})^{\frac{1}{\beta}-1} v_{nt} - v_{mt}]}{\alpha} \right)^{\frac{1}{\gamma-1}} & \text{for } m < \bar{m} \\ \left[\frac{1}{\alpha} (\lambda^{\frac{1}{\beta}-1} - 1) v_{\bar{m}t} \right]^{\frac{1}{\gamma-1}} & \text{for } m = \bar{m} \end{cases}. \tag{7}$$

Intuitively, firms choose a higher arrival rate of innovations when the cost of R&D α is low, and when the expected gain from innovating is high, captured by the

probability of moving to different positions in technology gap space upon innovating $\mathbb{F}_m(n)$, the value v_n of being at gap n , and the minimum size of quality improvements λ . All else equal, greater expected innovativeness of laggards (higher probability that they catch up to or overtake the leader), encourages more innovation by laggard firms. However, the v_n terms also capture the probability of being displaced in the future, so these values are endogenously determined along with the chance of displacement by rivals due to innovation or the chance of being hit with an exit shock δ_e . At t , the value of a potential entrant in product line i in sector j is simply $V_{ijt}^e = \delta_e V_{ijt}$.

3.4 Patent Quality: Model

The model analog of the patent quality measure in Section 2 is computed as follows. The patent quality measure captures the change in market value $V(m, q) = v_m q^{\frac{1}{\beta}-1}$ of a firm conditional on patenting relative to the pre-innovation value of the firm. If a firm at gap m with quality q innovates n steps, then patent quality is

$$\begin{aligned} \text{pat. qual.} &= \frac{V(m+n, \lambda^n q) - V(m, q)}{V(m, q)} \\ &= \frac{v_{m+n}}{v_m} (\lambda^n)^{\frac{1}{\beta}-1} - 1 \end{aligned} \quad (8)$$

Thus patent quality depends on the step size λ , the number of steps n , and the value function constants v_m which are endogenous to the market structure of the industry, the innovation technology of different firm types, and the risk of displacement by competitors, among others. Appendix B.10 derives the relationship between innovation rates and expected patent quality.

3.5 Equilibrium Output

Plugging in the intermediate goods firms' pricing decisions yields the following expression for final output Y_t , derived in Appendix B.3:

$$Y_t = \frac{1}{2} \frac{L}{1-\beta} P^{\frac{1-\beta}{\beta}} \sum_{m=-\bar{m}}^{\bar{m}} Q_{mt}, \quad (9)$$

where Q_{mt} is defined as:

$$Q_{m,t} = \begin{cases} (p_m^{1-\epsilon} + (\lambda^{-m})^{\epsilon-1} p_{-m}^{1-\epsilon} + (\zeta)^{\epsilon-1} \eta^{1-\epsilon})^{\frac{1-\beta}{\beta(\epsilon-1)}} \int_0^1 q_{i,t}^{\frac{1-\beta}{\beta}} \mathbb{1}_{\{i \in \mu_{mt}\}} di, & m \in \{-\bar{m}, \dots, 0\} \\ (p_m^{1-\epsilon} + (\lambda^{-m})^{\epsilon-1} p_{-m}^{1-\epsilon} + (\zeta \lambda^{-m})^{\epsilon-1} \eta^{1-\epsilon})^{\frac{1-\beta}{\beta(\epsilon-1)}} \int_0^1 q_{i,t}^{\frac{1-\beta}{\beta}} \mathbb{1}_{\{i \in \mu_{mt}\}} di, & m \in \{1, \dots, \bar{m}\} \end{cases}. \quad (10)$$

Here μ_{mt} is the measure of firms at each technology gap m at time t (normalizing the measure of large firms to one) and Q_{mt} is a particular index of the qualities of all firms at gap m . The change in output between t and $t + dt$ will therefore depend on the changes $\dot{Q}_{mt}$ for each technology gap m which in turn depend on the innovation arrival rates x_{mt} chosen by firms and the exogenous distribution of quality improvement sizes $\mathbb{F}(n)$. Let

$$P_m \equiv \begin{cases} (p_m^{1-\epsilon} + (\lambda^{-m})^{\epsilon-1} p_{-m}^{1-\epsilon} + (\zeta)^{\epsilon-1} \eta^{1-\epsilon})^{\frac{1-\beta}{\beta(\epsilon-1)}}, & m \in \{-\bar{m}, \dots, 0\} \\ (p_m^{1-\epsilon} + (\lambda^{-m})^{\epsilon-1} p_{-m}^{1-\epsilon} + (\zeta \lambda^{-m})^{\epsilon-1} \eta^{1-\epsilon})^{\frac{1-\beta}{\beta(\epsilon-1)}}, & m \in \{1, \dots, \bar{m}\} \end{cases}.$$

The P_m terms weight the change in qualities of firms at gap m depending on the prices set by firms at gap m and $-m$, capturing static distortions from firms' markups. Note that entry and exit are not a source of growth in the model because they have no impact on the qualities of the intermediate goods in the economy or on markups. The final component determining output will be the measure of firms at each technology gap μ_{mt} that is itself an endogenous object. The next section describes how to solve for the measures μ_{mt} .

3.6 Distribution Over Technology Gaps

Firms move to gap n through innovation from a lower technology gap m , or because their competitor innovates to gap $-n$. The distributions $\mathbb{F}_m(n)$ and $\mathbb{F}_{-m}(-n)$ respectively determine these probabilities, combined with the innovation efforts of firms at m and $-m$, for all $m < n$ and $-m < -n$. The outflows from gap n are due to a firm at n or its competitor at $-n$ innovating. Putting this together into the Kolmogorov forward equations for the evolution of the mass of firms at each gap:

$$\dot{\mu}_{nt} = \sum_{m=-\bar{m}}^{n-1} x_m \mathbb{F}_m(n) \mu_{mt} + \sum_{m=n+1}^{\bar{m}} x_{-m} \mathbb{F}_{-m}(-n) \mu_{mt} - (x_n + x_{-n}) \mu_{nt}. \quad (11)$$

The highest and lowest gaps are special cases because of spillovers: if the firm at the highest gap innovates both firms remain at the same gap:

$$\dot{\mu}_{-\bar{m}t} = \sum_{m=-\bar{m}+1}^{\bar{m}} x_{-m} \mathbb{F}_{-m}(\bar{m}) \mu_{mt} - x_{-\bar{m}} \mu_{-\bar{m}t} \quad (12)$$

$$\dot{\mu}_{\bar{m}t} = \sum_{m=-\bar{m}}^{\bar{m}-1} x_m \mathbb{F}_m(\bar{m}) \mu_{mt} - x_{-\bar{m}} \mu_{\bar{m}t}. \quad (13)$$

On a balanced growth path, $\mu_{mt} = \mu_m$ for all m, t . Replacing the left sides with zero change in equilibrium and the measures on the right sides with the constants μ_n, μ_m defines a system of $2\bar{m} + 1$ equations in $2\bar{m} + 1$ unknowns that determine the steady state distribution of firms over possible technology gaps. There are two additional restrictions on the solution to this system: (1) for each firm at m there is a firm at $-m$ (that is, the stationary distribution is symmetric); (2) I impose the restriction that the measure of all large firms sums to one.

3.7 Output Growth

Differentiating equation 9 with respect to time yields the following expression for the growth rate:

$$\frac{\dot{Y}_t}{Y_t} = g_{Yt} = \frac{1}{2} \frac{1}{1-\beta} \sum_{m=-\bar{m}}^{\bar{m}} \frac{\dot{Q}_{mt}}{Y_t}.$$

It's useful to define:

$$\tilde{Q}_{mt} = \int_0^1 q_{m,t,i}^{\frac{1-\beta}{\beta}} \mathbb{1}_{\{i \in \mu_{mt}\}} di \quad (14)$$

so that:

$$g_{Yt} = \frac{1}{2} \frac{1}{1-\beta} \sum_{m=-\bar{m}}^{\bar{m}} P_m \frac{\dot{\tilde{Q}}_{mt}}{Y_t}.$$

The subsequent analysis focuses on a balanced growth path where $\frac{\dot{\tilde{Q}}_{mt}}{Y_t}$ is constant for all m . On this balanced growth path consumption and output grow at a constant growth rate g and the mass of firms at each technology gap μ_m is constant. In

general it is not possible to solve for this growth rate in closed form, but for a given set of model parameters it is possible to check the existence and uniqueness of such a balanced growth path and find the value of g as the solution to a system of equations. A derivation of these results is provided in Appendix B.4.

3.8 Equilibrium Definition

Let $R_t = \int_0^1 \sum_{i=1}^2 R_{ijt} dj$ denote total research and development spending, C_t total consumption, and $K_t = \int_0^1 \eta(k_{1jt} + k_{2jt} + k_{fjt}) dj$ total purchases of final goods for production of intermediate goods. A Markov perfect equilibrium is an allocation $\{k_{ijt}, k_{fjt}, K_t, x_{ijt}, R_t, Y_t, C_t, L, \mu_{mt}, Q_{mt}, A_t\}_{i \in \{1,2\}, j \in [0,1], m \in [-\bar{m}, \bar{m}]}^{t \in (0, \infty)}$ and prices $\{r_t, W_t, p_{ijt}\}_{i \in \{1,2,f\}, j \in [0,1]}^{t \in (0, \infty)}$ such that for all t : (i) Households choose C_t and A_t to solve the problem in section 3.1. (ii) Final goods firms solve their problem to hire labor L and buy intermediate goods k_{ijt} and k_{fjt} optimally given the problem in section 3.2. (iii) Intermediate good firms choose p_{ijt} and x_{ijt} to solve their innovation and price-setting problems described in section 3.3. (iv) The final goods market clears: $Y_t = C_t + R_t + K_t$. (v) The asset market clears, pinning down r_t via the Euler equation. (vi) Labor market clears, pinning down W_t from the final good firm's problem. (vii) μ_{mt} and Q_{mt} are consistent with firms' choices of x_{ijt} .

4 Model Estimation

I estimate an initial steady state for the model by matching various moments for the U.S. economy in the period of high patent quality between 1994 and 2003 ("1990s"). Using this initial calibration I describe firms' pricing and innovation strategies to develop intuition about the model. I then re-estimate the model parameters for 2004-2017 ("2000s") to infer changes to the economy between these two periods.

4.1 Data for the Estimation

The data sources and computation methods for the data moments are described in detail in Appendix A and summarized here. Three aggregate moments are used as targets: the TFP growth rate (Comin et al. 2025), R&D as a share of GDP from the

OECD Main Science and Technology Indicators, and the profit share of GDP from the BEA. The other targets come from Compustat combined with [Kogan et al. \(2017\)](#)’s estimates of patent values. The Compustat sample covers around 14,000 publicly listed non-financial firms over this period in a wide variety of industries. Around 4,200 of these firms patented at least once between 1994 and 2017. In terms of representativeness, Compustat firms account for between 30 and 35% of all patents filed in the U.S. over this period. Manufacturing makes up around 80% of the patenting firm-year pairs and 50% of the firm-year pairs overall.

4.2 Baseline Calibration for the 1990s

Four parameters are calibrated outside the model using standard values. The inverse intertemporal elasticity of substitution ψ is set to 1. The labor share, β , is set to 0.6. This implies an elasticity of substitution across sectors of $\frac{1}{\beta} = \frac{5}{3}$, within the range of upper-level elasticities of substitution estimated in [Hobjin and Nechio \(2019\)](#). The curvature of the R&D cost function, γ , is chosen to match empirical evidence on the elasticity of patenting to R&D expenditures discussed in [Acemoglu et al. \(2018\)](#). The maximal technology gap $\bar{m}$ is set to 16.

The rest of the parameters for the baseline model, shown in Table 1, are estimated using a simulated method of moments approach described in Appendix C.1 to match targets for the 1990s equilibrium. Appendix C.2 describes the solution method for finding the model steady state.²⁰ The fringe quality distance ζ is adjusted such that, given the estimated ϵ , the incumbents each have a market share of 30% in the neck and neck state. The leader’s market share then increases from 30% as their technology gap grows.

The targets are listed in Table 2 and include the main phenomena of interest: aggregate productivity growth, concentration (measured as the average market leader’s share of industry sales), plus the profit share, the R&D share of GDP, the rate of leadership turnover from either the exit shock or being overtaken by the other large firm, and patent quality heterogeneity for leaders and followers, plus the standard

²⁰The key determinant of the time to solve the model computationally is the number of step sizes $\bar{m}$. Given this choice, I let one patent in the model be equivalent to 10 patents in the data, scaling up the three patent quality targets.

	Value	Meaning/source
ψ	1	Inverse intertemporal elasticity of substitution
ρ	0.040	Rate of time preference (annual)
β	0.6	Labor share/Nechio & Hobijn (2019)
ϵ	4.999	Elasticity of substitution within sectors
η	0.802	Marginal cost of intermediate producers
δ_e	0.094	Exogenous entry/exit rate (annual)
λ	1.048	Min. qual. improvement
γ	2	Curvature of R&D function
α	2.000	R&D cost parameter
ζ	0.767	Fringe quality distance
$\bar{m}$	16	Maximum number of steps ahead
ϕ	0.985	Shape of idea quality distribution

Table 1: Model parameters, 1990s steady state. Estimated parameters in bold. Estimation details in the text and in Appendix C.1.

deviation of patent quality. Including the average patent quality of market leaders and followers separately helps identify λ and ϕ individually, as does the standard deviation of patent quality. R&D as a share of output is included to discipline the R&D cost parameter. The discount rate ρ also affects the aggregate level of R&D investment. The entry/exit shock δ_e helps match leadership turnover. The competition parameter ϵ moves concentration and the profit share in opposite directions, capturing the conflicting effects of the rise of superstar firms and rising market power.

The model performs well in fitting the targeted moments, though it slightly underestimates the patent quality differences between leaders and followers. Despite this, Figure 4 compares R&D intensity (R&D/sales) in the data and the model (which is not targeted), and the model is able to reproduce the sharply decreasing relationship between a firm’s market share and its R&D intensity.

The estimated parameters are reasonable and match non-targeted moments for the U.S.: a discount rate ρ of 4% implies an annual real interest rate of 5.6%. An elasticity of substitution ϵ of 4 results in a sale-weighted average markup of 1.33, close to the value of 1.27 in the data (Table 3). The entry/exit rate δ_e of about 9.4%

	Data	Model
TFP growth, %	1.57	1.58
Avg. leader market share, %	43.23	42.62
R&D share of GDP, %	1.80	1.76
Profit share of GDP, %	7.71	7.11
Leadership turnover, %	13.74	14.16
Patent quality, avg., followers, %	19.02	16.67
Patent quality, avg., leaders, %	6.87	8.44
Patent quality, std. dev., %	25.75	27.61

Table 2: Model fit for targeted moments from estimation of 7 parameters for the 1990s. Estimation details in the text and in Appendix C.1. The method for computing each moment in the data is described in Appendix A.2.

per year is in line with entry and exit rates for the U.S. reported by [Decker et al. \(2016\)](#). The estimated model also roughly matches the combined market share of the two largest firms, the leader’s share of industry profits, and the fact that market leaders generate more market value of patents per R&D dollar spent than followers despite having lower patent quality (Table 3). The model also roughly matches the standard deviation of productivity growth rates for followers who patent (Table 3). Changes in the latter moment are useful for distinguishing between the declining idea quality hypothesis from declining knowledge diffusion (see Appendix A.6.1 for further discussion).

The markups and market shares for quality leaders as a function of the leader’s technology gap are plotted in Figure 5. The leader’s market share and markup rise as the technology gap widens (that is, as the leader’s relatively quality improves).

4.3 Re-estimation for the 2000s

Re-estimating the model for the 2000s uses the model to infer the role of different channels suggested in the literature to explain changes in productivity growth, patent quality, and concentration and compare the importance of these other channels to the importance of the idea quality channel. Changing the discount rate ρ captures

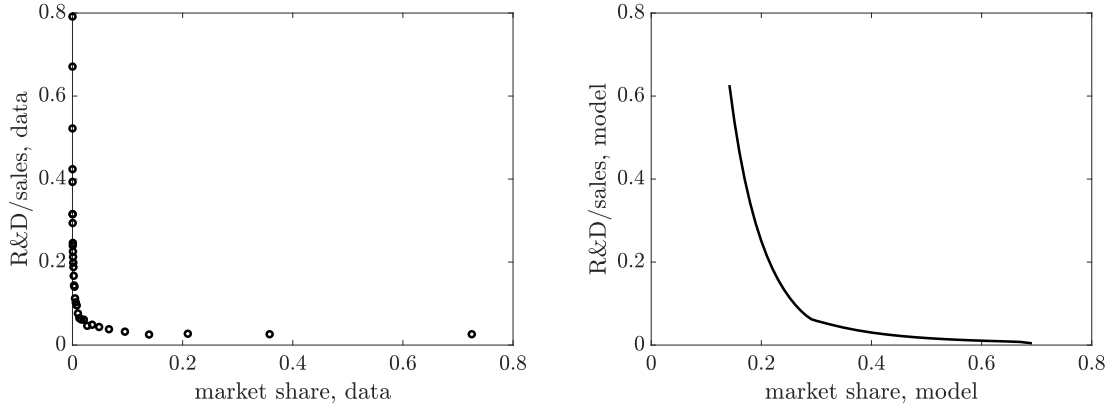


Figure 4: Model’s fit for non-targeted R&D intensity as a function of market share in the data (left panel) and the model (right panel). Data is a binned scatterplot of $XRD/SALE$, winsorized at the 10% level, for the years 1994-2003.

	D:1990s	M:1990s
Sale-weighted markup, %	27.47	32.58
Top two firms market share, %	70.48	67.73
Leaders share of industry profits, %	60.95	72.31
Pat. val. per R&D, leaders	10.08	26.19
Pat. val. per R&D, followers	3.98	2.29
Std. dev. TFPR growth, followers w/ pat.	14.50	9.80

Table 3: Model fit for non-targeted moments in the 1990s steady state. Appendix A.2 contains more details on the calculation and data sources.

the interest rate channel proposed by [Liu, Mian, and Sufi \(2021\)](#).²¹ A decrease in incumbent exit shocks δ_e can capture declining knowledge diffusion from incumbent firms to new entrants or rising entry costs ([Akcigit and Ates \(2021\)](#), [Corhay, Kung, and Schmid \(2025\)](#)). An increase in the research cost parameter α implies that more R&D spending is needed to achieve the same arrival rate of innovations, capturing the cost side of the hypothesis of [Bloom et al. \(2020\)](#) that ideas are getting harder to find.

²¹For the estimated model, even at very low interest rates (for example 0.1%) the strategic effects described in [Liu, Mian, and Sufi \(2021\)](#) do not dominate the first-order effect of lowering the interest rate on firms’ R&D. See [Goldberg, Lopez-Salido, and Chikis \(2021\)](#) for further discussion.

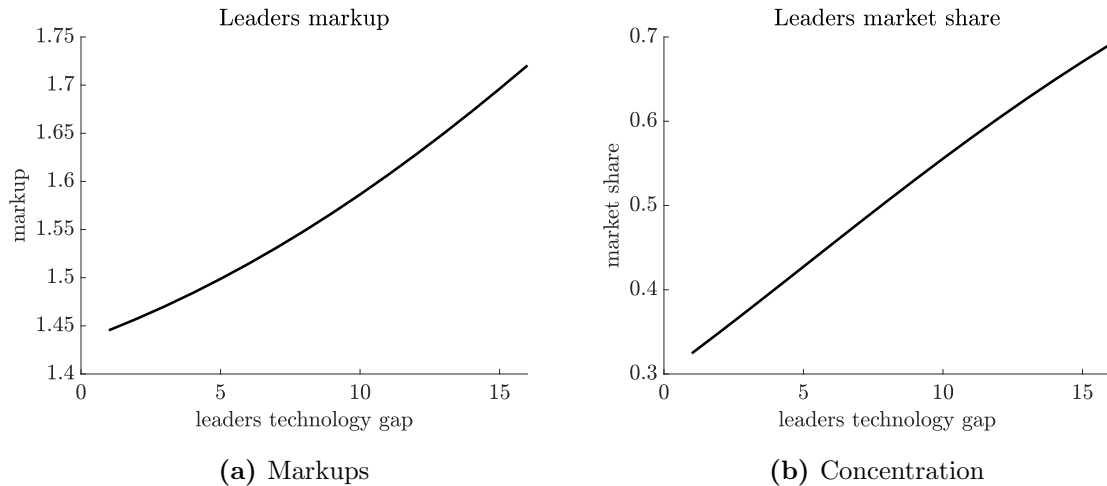


Figure 5: Leader’s markup (price over marginal cost) and market share as a function of the leader’s technology gap m in the model using the 1990s calibration in Table 1.

A decrease in the elasticity of substitution within sectors ϵ captures increased market power. On the other hand, an increase in ϵ captures the superstar firm hypothesis of Autor et al. (2020) that competitive pressures within industries have risen, causing the most productive firms to capture a larger share of total industry sales. Finally, changes in λ and ϕ govern the expected quality of new ideas for different types of firms by changing the distributions $\mathbb{F}_m(n)$. In particular rising ϕ captures heterogeneity in the decline in the quality of new ideas for leaders and laggards.

Table 4 shows the model fit for the 2000s estimation with the model also matching the 2000s moments well. In the data, productivity growth slowed substantially and the average market leader’s sales share grew by about five percentage points. Aggregate R&D expenditures as a share of GDP grew (Bloom et al. 2020). Despite rising R&D expenditures, the profit share also grew. Here the model struggles slightly to match the rise in the profit share, R&D expenditures, and concentration simultaneously, but still implies a rise in the profit share relative to the 1990s model steady state. The implied declines in patent quality are 44% (vs. 38% in the data) for laggards and 12% decline for leaders (vs. 29% in the data). Later robustness explores an alternative calibration that matches the changes in the patent quality moments exactly.

Table 5 compares the estimated parameters for the two steady states. Consistent with Akcigit and Ates (2023) and Decker et al. (2016), the rate δ_e at which incumbents

	Data	Model
TFP growth, %	0.66	0.66
Avg. leader market share, %	48.12	48.12
R&D share of GDP, %	1.89	1.92
Profit share of GDP, %	9.20	7.43
Leadership turnover, %	9.28	9.96
Patent quality, avg., followers, %	11.86	9.49
Patent quality, avg., leaders, %	4.91	7.43
Patent quality, std. dev., %	19.49	19.81

Table 4: Model fit for targeted moments from estimation of 7 parameters for the 2000s. Estimation details in the text and in Appendix C.1. The method for computing each moment in the data is described in Appendix A.2.

are exogenously displaced by new entrants declines. Generating new ideas becomes more expensive (α increases) and the quality of new ideas worsens,²² consistent with Bloom et al. (2020) and the evidence in Section 2. To match the rise in concentration, competition ϵ increases. The effects of each of these changes on growth, concentration, and other moments are decomposed in Section 5.

4.4 Additional Model Validation

Appendix Table A.4 shows the model fit for changes in additional non-targeted moments. The 2000s model steady state replicates, without targeting, rising productivity differences, markups, a decline in TFP growth dispersion for followers who patent, and a rise in industry leaders' share of total R&D expenditures.

The model performs well in matching not just the average level of concentration but the entire (non-targeted) distribution of leaders' market shares across sectors of the economy in both periods (Figure 6). The shift is mainly due to increased average quality differences between leaders and followers, consistent with the findings of Andrews, Criscuolo, and Gal (2015) and Andrews, Criscuolo, and Gal (2016) that

²²Despite the slight rise in the minimum step λ , the new idea quality distribution implies a decline in the average quality improvement from 18% to 9%, holding fixed the distribution of firms over technology gaps and the innovation rates.

	1990s	2000s	Meaning
ρ	0.040	0.059	Rate of time preference (annual)
ϵ	4.999	4.135	Elasticity of substitution within sectors
η	0.802	1.222	Marginal cost of intermediate producers
δ_e	0.094	0.085	Exogenous entry/exit rate (annual)
λ	1.0481	1.0525	Min. qual. improvement
α	2.000	2.321	R&D cost parameter
ϕ	0.985	1.721	Shape of idea quality distribution

Table 5: Comparison of estimated parameters, 1990s vs. 2000s model steady states. Estimation details in the text and in Appendix C.1.

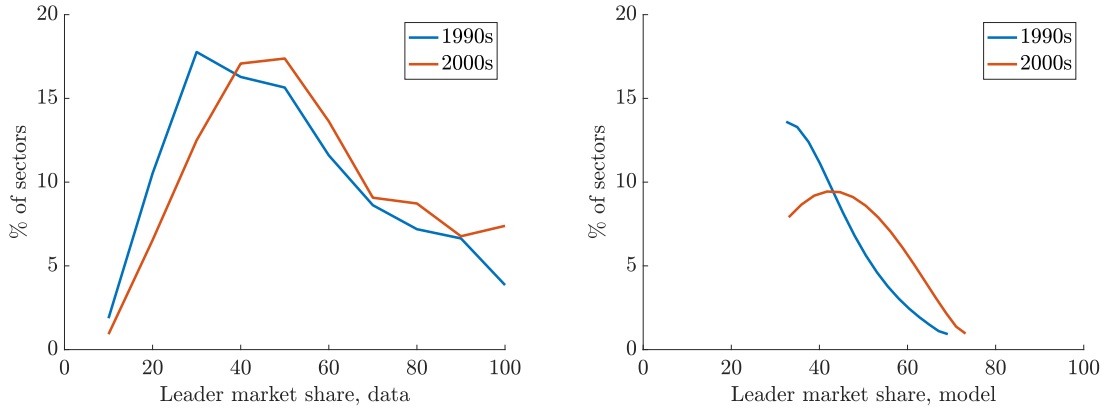


Figure 6: Shift in concentration across sectors, data vs. model. Leader market share is for 4-digit SIC industries in Compustat. See Appendix A.1 for a description of the sample.

productivity differences within industries have grown. This divergence is particularly pronounced in ICT intensive sectors, and sectors with wider productivity gaps have experienced deeper productivity slowdowns. Figure A.8 shows that rising concentration and productivity slowdowns are correlated at the sector level.

The model predicts that the average leader's share of total industry R&D rises from 30% to 58%. Among public firms, market leaders now perform a larger share of industry R&D than in the 1990s and capture a larger share of industry patents (Figure A.7), though the increase from a low of 38% in the late 1990s to a peak of 50% in 2010 is smaller than in the model. R&D in Compustat includes purchased R&D but

not acquisitions and purchases of intangibles like patents, which have risen for large firms over this period (Crouzet and Eberly 2019) and may explain the discrepancy. Anderson and Kindlon (2019) find a decline in R&D intensity among companies with fewer than 250 employees and an increase for larger firms in the National Science Foundation’s Business R&D and Innovation Survey covering public and private firms. Akcigit and Ates (2023) document increasing concentration of patents among the top 1% of patenting firms and increasing flows of R&D workers from small to large firms, all of which are consistent with a shift in R&D activity toward large firms.

5 Results

5.1 Role of Declining Idea Quality

Table 6 summarizes the role of the estimated changes in λ and ϕ compared to changes in all the other parameters at once to match the moments of interest. To decompose the effect of a change in the quality improvement distribution, I compute the share of the changes in the data that are explained by changing λ and ϕ as follows:

$$\frac{M_j(\boldsymbol{\theta}_{1990s}, \lambda_{2000s}, \phi_{2000s}) - M_j(\boldsymbol{\theta}_{1990s}, \lambda_{1990s}, \phi_{1990s})}{D_{j,2000s} - D_{j,1990s}} \times 100, \quad (15)$$

where M_j is steady state moment j as a function of the model parameters and $\boldsymbol{\theta}$ is the vector of other model parameters $(\rho, \epsilon, \eta, \delta_e, \alpha)$. $D_{j,t}$ denotes the moment’s value in the data at time $t \in \{1990s, 2000s\}$.²³

The estimated changes in the distribution of quality improvements alone explain 113% of the productivity slowdown and 89% of the rise in concentration between the two steady states, though I later show the latter effect is slow to materialize. Changes in idea quality also explain most (93%) of the rise in the profit share and three quarters of the decline in leadership turnover.

Turning to the mechanisms, in the counterfactual with lower idea quality laggard firms respond by choosing a lower innovation rate x (Figure 7a).²⁴ The effect on

²³Note, however, that these are not the marginal effects of the parameters on each moment; the moments are endogenously determined in steady state.

²⁴Figure B.4 shows the impact of the estimated changes in the idea quality parameters on expected idea quality and expected patent quality for each firm type holding other parameters fixed. Only

<u>Moment</u>	<u>Data</u> change (pp)	<u>Decomposition</u>	
		Innov. (%)	All others (%)
TFP growth	-0.9	112.7	-30.7
Leader market share	4.9	88.6	15.8
R&D/GDP	0.1	-972.2	2161.3
Profits/GDP	1.5	93.0	-139.2
Leadership turnover	-4.5	76.9	5.7
Pat. qual., avg., followers	-7.2	103.1	1.0
Pat. qual., avg., leaders	-2.0	58.4	-6.7
Pat. qual., std. dev., %	-6.3	134.3	-2.2

Table 6: Share of changes in each data moment between 1990s and 2000s explained by the estimated parameter changes in Table 5. Column labelled “Innov.” holds other parameters fixed at 1990s values while λ and ϕ change to their estimated 2000s values. “All others” column holds λ and ϕ fixed and allows the five other estimated parameters to change to 2000s values. Positive sign in the second and third columns indicates same direction of change as in the data. See text for more details.

leaders’ innovation policies is ambiguous ex ante. Most leaders face lower expected patent quality and all leaders are less likely to be overtaken so face less need to escape competition through innovation and leaders face lower idea and patent quality. Both of these forces lower desired innovation rates. On the other hand, facing a lower probability of being overtaken, leaders discount future gains from innovation less and the value of attaining market leadership is higher. In the calibrated model the latter forces dominate and far-ahead leaders choose a higher innovation rate.

Together, these responses cause the economy’s distribution of leader market shares to shift right: more sectors feature a dominant leader that has high relative product quality (Figure 7b). The increases in concentration, the average sale-weighted markup (from 33% to 35%), and the profit share in the equilibrium with lower idea quality are driven by this composition effect: fixing innovation rates across firm types at their 1990s values but changing the idea quality distribution would result in a *tighter* 5% of firms, the largest leaders, have increased expected idea quality. 15% have higher expected patent quality.

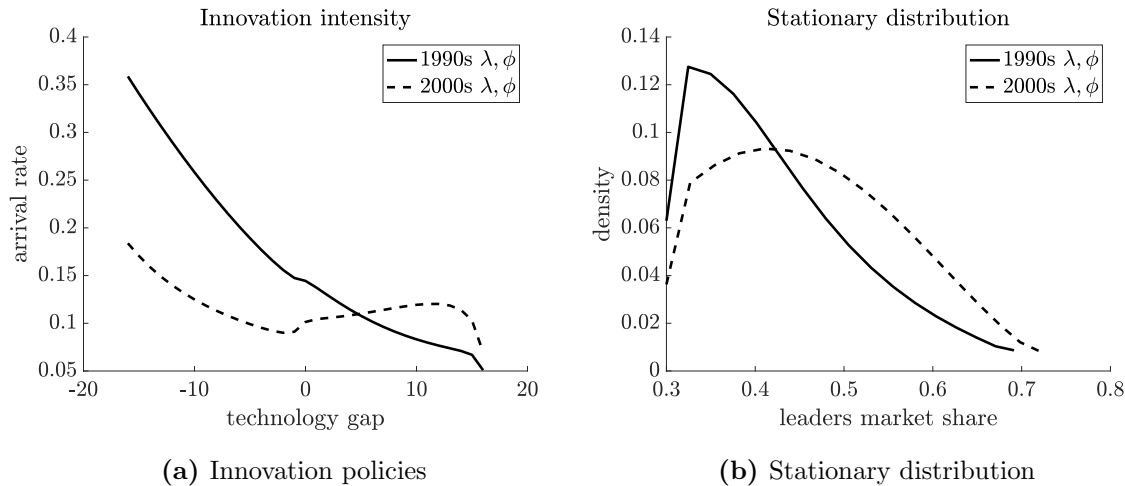


Figure 7: Innovation policies x as a function of the technology gap for all firms (panel a) and the distribution of sectors over the leader’s market share (panel b) comparing the 1990s steady state (solid line) in Table 1 to a counterfactual equilibrium where other parameters are fixed at 1990s values and ϕ and λ change to their 2000s value (dotted line).

distribution around the neck-and-neck state, because competitors pull away from each other less frequently under a higher ϕ (Figure B.5).

The growth rate declines when idea quality is lower for two reasons. First, there is the endogenous strategic effect that comes from changes in firms’ innovation policies x (Figure 7a). Leaders’ improvements are more incremental on average, so both the level effect of reduced R&D and the reallocation effect contribute to slower productivity growth. Second, reducing the average size of quality improvements obviously exogenously lowers the growth rate.

Two different decompositions, formally described in Appendix B.5, can be used to quantify the relative importance of these two channels. First, to isolate the strategic effect, I solve for the economy’s growth rate holding the innovation parameters λ and ϕ fixed but varying firms’ innovation policies x to their values in the 2000s counterfactual where λ and ϕ change and other parameters are fixed. These changes in effort alone can explain 49% of the productivity slowdown. The other decomposition isolates the first order effect by fixing firm innovation policies to the 1990s model steady state and reducing idea quality exogenously. This alone can account for 85% of the productivity slowdown in the model. The fact that the two sum to more than 100%

of the slowdown is due to interactions between the two in steady state. The strategic effect can therefore be said to account for between 15%-49% of the slowdown.

Declining Idea Quality: Transition Dynamics. To understand the dynamics of the transition from the high to low idea quality equilibrium, I consider a permanent decrease in idea quality given by the changes in λ and ϕ estimated in Table 5. The idea quality parameters change smoothly for the first three years and then remain permanently at their new steady state values. After the initial surprise, firms anticipate the path of expected quality improvements over the transition. Computational details are in Appendix C.3.

Figure 8 shows the evolution of productivity growth and concentration. After an initial boom, productivity growth closely tracks the decline in idea quality. The initial boom in productivity growth is due to a boom in R&D investment when the transition begins. The boom occurs because firms have perfect foresight that idea quality is declining and that the terminal value of market leadership is higher than in the initial steady state. Concentration rises slowly and after 25 years has not yet reached its new steady state value. Innovations are somewhat infrequent, so it takes time for leaders to pull ahead and laggards to fall as far behind as they are in the new steady state on average. The economy does eventually converge to the new steady state (Appendix Figure B.7). The effect of idea quality on concentration after 25 years explains about 44% of the increase in the data, compared to 89% when comparing steady states. These patterns are consistent with Figure 1 which shows the TFP growth rate boomed in the 1990s before declining quickly in the early 2000s and then remaining roughly flat, while concentration continued to rise through 2017.

Robustness. The effects of the declining idea quality on productivity growth and concentration are qualitatively robust to several modifications. First, Appendix B.6 recomputes the changes due to the decline in idea quality in the model without the competitive fringe where the two incumbents compete a la Bertrand. The level and change in productivity growth rates are roughly the same in that alternative model. The level of concentration is higher and idea quality changes explain 40% rather than 88% of the increase in data. The results also do not depend on the randomness of innovation step sizes (Appendix B.7).

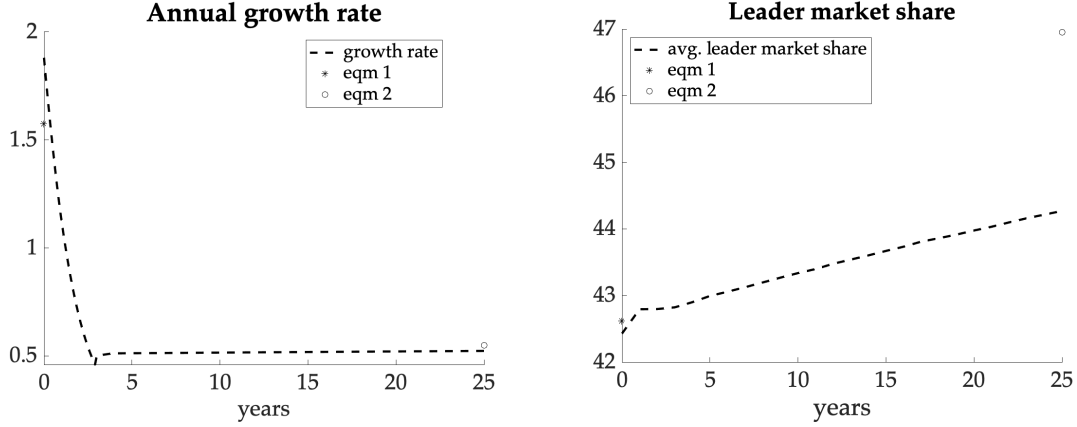


Figure 8: Transition dynamics for productivity growth and concentration over the transition to a lower level of idea quality. Details in the text and Appendix C.3.

The results for concentration do depend on the estimated changes in the idea quality parameters λ and ϕ . If λ and ϕ are calibrated to exactly match the percent decline in the patent quality moments for leaders and followers, λ is estimated to decline and ϕ does not increase as much as in the main exercise (Appendix B.8). This generates a similarly sized productivity slowdown and boom and bust over the transition as the baseline changes to idea quality, but concentration falls slightly. In spite of this the profit share rises, consistent with the data. The main exercise infers changes in idea quality to jointly fit all the moments, including concentration, and the patent quality data does not offer a way to measure idea quality directly. A fairly large degree of heterogeneity in the decline in idea quality across large and small firms is needed to increase concentration.²⁵ Together, the transition results and these robustness exercises show that while changes in idea quality can increase concentration, they do not necessarily do so. The best fit for the moments of interest in the 2000s from the main exercise imply sufficiently large heterogeneity in idea quality to generate a rise in concentration.

²⁵Appendix B.9 cuts the step size λ in half but keeps the set of firm types fixed by doubling $\bar{m}$. This does not increase concentration and does not affect the standard deviation of patent quality.

<u>Moment</u>	<u>Model</u>			
	$\epsilon = 3.0$	$\epsilon = 4.0$	$\epsilon = 4.1$	$\epsilon = 5.0$
TFP growth, %	1.78	1.58	1.56	1.52
Avg. leader market share, %	36.92	42.62	43.37	48.13
R&D share of GDP, %	2.10	1.76	1.74	1.68
Profit share of GDP, %	7.70	7.11	7.06	6.83
Leadership turnover, %	14.78	14.16	14.11	13.91
Patent qual., avg., followers, %	13.32	16.67	17.15	20.26
Patent qual., avg., leaders, %	6.81	8.44	8.64	10.12
Patent qual., std. dev., %	21.42	27.61	28.52	34.65
Sale-weighted markup, %	40.02	32.58	32.05	29.69

Table 7: Steady state comparison, role of elasticity of substitution ϵ , holding all other parameters fixed at the 1990s values in Table 1. $\epsilon = 3.0$ is the market power experiment, $\epsilon = 4.0$ is the estimated 1990s value, $\epsilon = 4.1$ is the jointly estimated 2000s value, and $\epsilon = 5.0$ is the superstar firm experiment where ϵ changes to match concentration in the 2000s.

5.2 Increasing Market Power or Superstar Firms?

The main exercise in Section 4.3 estimates a slight rise in the competition parameter ϵ from 4 to 4.13, consistent with the “superstar firm” hypothesis of Autor et al. (2017) that competition within sectors has increased. In their static model, an increase in substitutability causes a reallocation of sales to the more productive “superstar” firms, raising measured sectoral TFP.²⁶ Dynamically, however, firms’ innovation policies will respond to changes in competition and the present model offers a way to study the growth effects of changing competition.

Using the same decomposition approach of fixing other parameters and varying ϵ , the estimated increase in competition produces a small productivity slowdown (about 2% of the entire productivity slowdown in the data) and explains about 15%

²⁶The model in Autor et al. (2020) generalizes this formulation. The exercise presented here studies the specific shock to market toughness suggested in Autor et al. (2017). This shock could represent more fierce import competition from abroad, particularly China, in recent years. Amiti and Heise (2024) find support for this in Census data on U.S. and foreign manufacturing firms.

of the rise in concentration (Table 7).²⁷ The rise in concentration is due to two forces. First, the static reallocation force operates: even if technology gaps were unchanged, these same gaps would generate a higher average leader market share. Second, increasing competition reduces the innovation effort of laggards (Figure B.6a) causing the average technology gap to grow (Figure B.6b) and growth to slow down.

The reason the joint estimation does not favor increased competition as the main driver of rising concentration is that it gives counterfactual predictions for the profit share, highlighting the central tension in using models with CES demand to explain both rising concentration and markups using a change in competition. A second reason is that higher values of ϵ raise patent quality in the model, contrary to the data. Patent quality (equation 8) measures the percent change in the value of the firm from innovation and is independent of the level of firm value. When ϵ is high, the change in firm value for the same product quality improvement is higher because markups and sales respond more to a change in the quality gap between leader and follower. Appendix B.10 derives an expression for firms' innovation policies x_m and shows they depend on expected patent quality, which rises, and on firm's current strategic value v_m . In the model these values v_m fall for all m when competition increases because markups are lower (Figure B.3). The net effect is a decline in innovation effort for laggard firms despite higher patent quality.

For the values considered, the effect of ϵ on the model moments is monotonic: the “market power” experiment in column 1 of Table 7 shows that lower ϵ increases steady state growth and reduces concentration, but has the same direction of change in patent quality, profits, and markups as the data. A larger change in ϵ , calibrated to match the entire rise in concentration, is shown in the last column of Table 7.

Rise of Superstar Firms: Transition Dynamics. The transition dynamics associated with an increase in competition ϵ to match the rise in concentration are plotted in Appendix Figures B.8. When competition increases, growth drops immediately and then recovers to the new steady state level, contrary to the boom seen in the late 1990s. Concentration rises quickly because increased competition results in reallocation of sales to market leaders even without any changes in technology gaps.

²⁷If ϵ changes to match the entire rise in concentration in the data, implying $\epsilon = 5$ (last column of Table 7), the associated productivity slowdown is 6% of the productivity slowdown in the data.

Robustness. The superstar firm experiment creates a similarly sized productivity growth slowdown of 0.1 percentage point in the model without the fringe as with the fringe (Appendix Table B.2). The effect on concentration is smaller in the model without the fringe (a 2 percentage point increase rather than 5).

5.3 Other Hypotheses

In addition to Table 6 showing the effects of all parameter changes besides the idea quality parameters at once, Table C.1 in Appendix C.4 reports the effect of changing each parameter individually from its 1990s value to its 2000s value. None of the other estimated parameter changes significantly reduce productivity growth and increase concentration simultaneously. The rise in the innovation cost parameter (ideas getting more expensive) α explains around 20% of the productivity slowdown. The estimated decline in knowledge diffusion raises the growth rate slightly because the value of incumbents v_m rises and it does not affect concentration. The knowledge diffusion mechanism presented in Akcigit and Ates (2023) predicts that the mean and standard deviation of patent quality and TFP growth conditional on patenting should rise, contrary to the data, see Appendix A.6.1 for further discussion.

6 Conclusion

New patents had less impact on growth in firm valuations in the 2000s compared to the 1990s. Smaller firms drove the boom in patent quality that has been attributed to the arrival of information technology as a general purpose technology, as well as the bust that began in the early 2000s. This finding contributes to the debate on whether ideas are getting harder to find, emphasizing heterogeneity across firms. Further work should investigate heterogeneity in the complementarity of general purpose technologies with firms' R&D investments, and determine whether similar patterns of idea quality and firm dynamics were present in the wake of previous technology waves.

To understand the consequences of this empirical fact I develop a general equilibrium growth model of innovations where multiple firms are active in each sector in each period. The model, estimated for the U.S. in the two different periods, points to smaller quality improvements as the main driver of the productivity slowdown. This

finding is robust to changing the exact nature of the industry structure or alternative estimation targets. The model suggests that declining idea quality is amplified by firms' endogenous responses: between 15-50% of the productivity slowdown in the model from declining idea quality is due to changes in firms' innovation rates.

The estimated changes in idea quality also produce a large increase in concentration, though computation of the transition dynamics reveals that these effects take a long time to materialize. I conclude that changes in idea quality can explain less than half of the rise in concentration in the data over the period of study and alternative explanations such as increased competition are needed to explain the rise in concentration over this period. A rise in the elasticity of substitution has the potential to explain rising concentration and the productivity slowdown, providing a dynamic complement to the experiment in [Autor et al. \(2017\)](#), one that rationalizes the emergence of superstar firms alongside a productivity slowdown for a standard Schumpeterian reason: firms' incentives to innovate fall because the value of market leadership is lower. However, this is inconsistent with patterns of markups and the profit share, which have risen over this period, so it cannot be the whole story.

A final area for future research is how policy can spur the development of new general purpose technologies. Past work suggests this is difficult because there are significant positive externalities for other sectors that the inventor of the general purpose technology does not internalize. This paper suggests there may also be winners and losers within other industries, further complicating this problem.

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A Online Data Appendix

A.1 Data Description

The firm-level panel data comes from Compustat, keeping only firms incorporated in the U.S. reporting in U.S. dollars. The sample excludes financial firms (SIC codes beginning with 6), utilities (SIC codes beginning with 49), agricultural firms (those with SIC<1000), and acquisition-related non-operational companies (SIC 9995). All firms in the sample are publicly listed. There are 14,249 unique firms in the sample from 1994-2017, and 134,733 firm-year pairs. 4,298 of these firms ever patented, and around 20% of the firms in the sample patent in a given year. Public firms captured a rising share of total patents filed in the U.S. over this period, rising from around 30% to 35%. Manufacturing firms are around 50% of firm-year pairs in the Compustat data but account for 80% of patenting year pairs. Services firms (those with SIC beginning with 7) account for 19% of firm-year pairs and 11% of patenting firm-year pairs. Other data sources and construction details for the targeted moments used in the paper are listed in Table A.1 and Table A.2 provides the same information for additional non-targeted moments.

A.2 Data Sources and Moment Computations

Table A.1 lists the source and, where necessary, computation method for each target moment from the data.

Targeted Moment	Source	Computation/Series Name
TFP growth	Comin et al. (2025)	Utilization-adjusted annual total factor productivity growth estimates allowing for non-zero profits
Leader market share	Compustat	Average of sales share (SALE), largest firm in each 4-digit SIC industry, weighted by industry size in terms of sales
R&D share of GDP	OECD Main Science and Technology Indicators	Business Expense R&D (private)/GDP
Profit share of GDP	Bureau of Economic Analysis/FRED	Profits with inventory valuation and capital consumption adjustments/Gross domestic income
Leadership turnover	Compustat	Share of 4-digit SIC industries with new sales leader per year
Patent quality mean and standard deviation	Kogan et al. (2017) /Compustat	$\Theta_{f,t}^{sm}$ is the market value of firm f 's patents issued in year t as in Kogan et al. (2017) (p. 688). $\text{pat. qual.}_{f,t} = \frac{\Theta_{f,t}^{sm}}{fNpats_{f,t}}$; where $M_{f,t-1}$ is market capitalization in the previous year PRCC.C*CSHO and $fNpats_{f,t}$ is the number of patents issued to firm f in year t . Drop observations over 100% (12 total).

Table A.1: Data sources and computation method for targeted moments used in the estimation.

Non-targeted moment	Source	Computation/Series Name
Citation-based patent quality	Kogan et al. (2017)	5-year forward citations Θ^{cw} received by firm f 's patents issued in year t $psgpp_{f,t} = \frac{\Theta^{cw}_{f,t}}{\sum_{s=1}^{t-1} \Theta^{cw}_{f,s}}$; where s is f 's first year in Compustat. Drop those with quality > 100%
Dollar value per patent	Kogan et al. (2017)	Mean patent value in dollar terms deflated by GDP deflator from BEA. Same sample selection as patent quality measures.
Pat. val. per R&D	Kogan et al. (2017)/Compustat	Θ_{sm} divided by lagged Compustat R&D expenditures XRD, conditional on patenting. Same sample selection as patent quality measures.
Ann. pat. growth	Kogan et al. (2017)/Compustat	Θ_{sm} divided by lagged Compustat market capitalization PRCC_C*CSHO, conditional on patenting. Same sample selection as patent quality measures.
Sale-weighted markup	Compustat	Markups estimated using the procedure in Appendix A.4 and aggregated using economy-wide shares of SALE.
Top two firms market share	Compustat	Sum of largest (by sales) two firms sales shares (SALE) within each 4-digit industry, averaged across industries.
Std. dev. TFPR growth, followers w/ pat.	Compustat	TFPR is estimated using the procedure in Appendix A.4. Winsorize top and bottom 5% of TFPR growth rates conditional on patenting then compute standard deviation.
Productivity gap	Compustat	TFPR is estimated using the procedure in Appendix A.4. $\log(TFP_{leader}) - \log(TFP_i)$ averaged across all firms i .
Leader share of R&D (or patents)	Compustat	Sum XRD (patents) within each 4-digit SIC and compute the largest firm's (by sales) share, averaging across industries.

Table A.2: Data sources and computation method for non-targeted moments reported in the paper.

A.3 Additional Patent Quality Figures

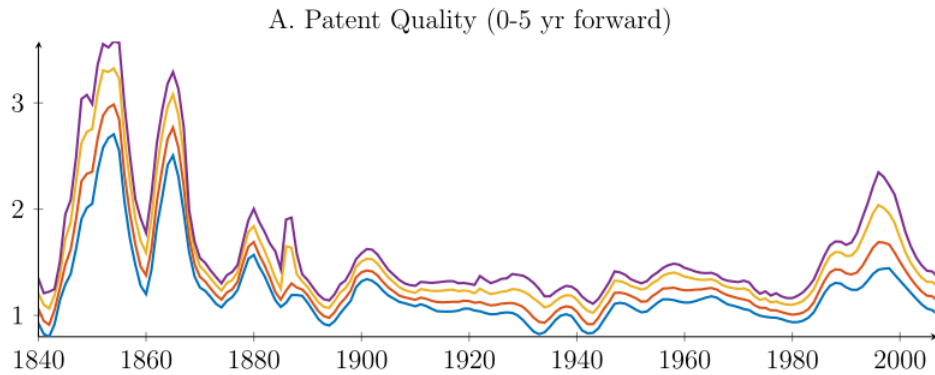


Figure A.1: Percentiles of text-based patent quality distribution over time. Blue = P50, Red = P75, Yellow = P90, Purple = P95. Source: [Kelly et al. \(2018\)](#) Figure 3a.

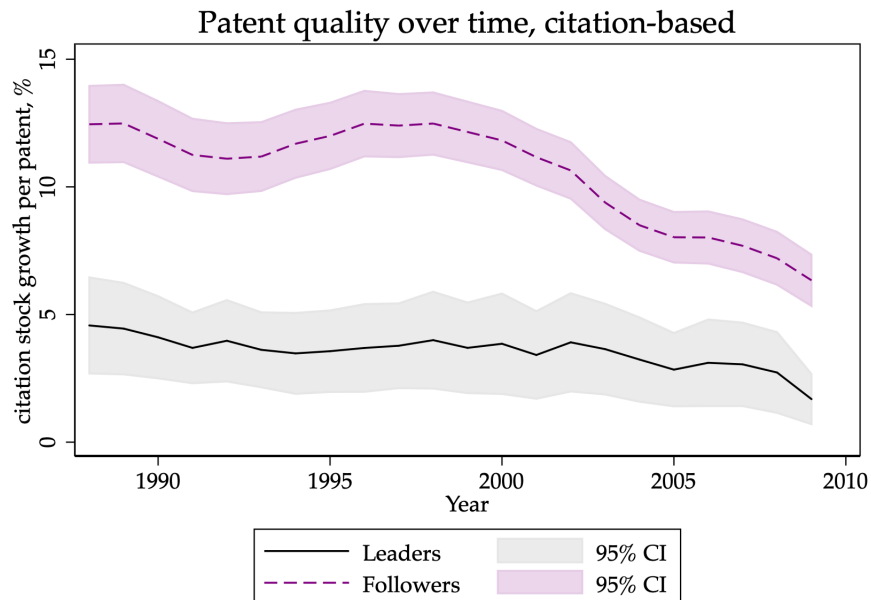


Figure A.2: Contribution of average new patent to firm's existing stock of patent citations (cumulated 5-year forward citations from [Kogan et al. \(2017\)](#)). Leader indicates sales leaders in 4-digit SIC industries in a given year and followers are all other public firms in the sample. Three year moving averages. See Appendix A.1 for more details.

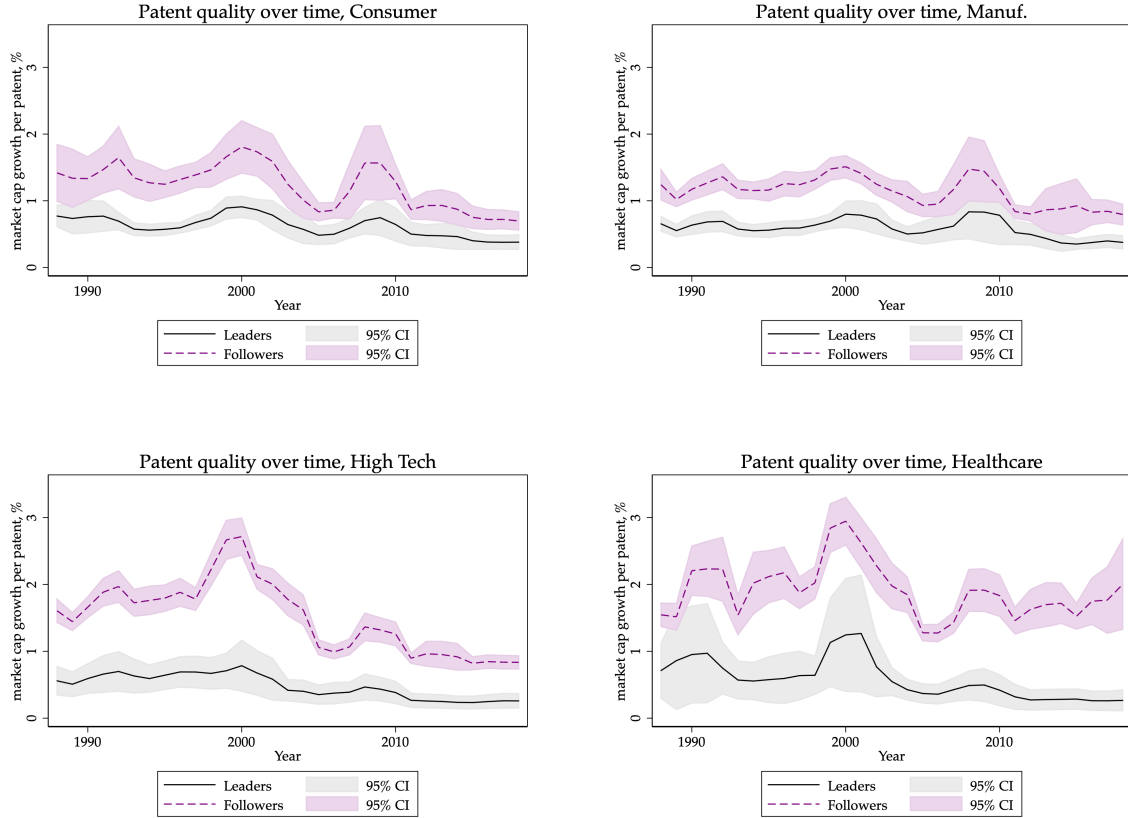


Figure A.3: Average patent quality differences between leaders and followers in Fama-French 5 broad industry categories, using estimated patent values from [Kogan et al. \(2017\)](#). Leader indicates sales leaders in 4-digit SIC industries in a given year and followers are all other public firms in the sample. Three year moving averages. See Appendix A.1 for more details.

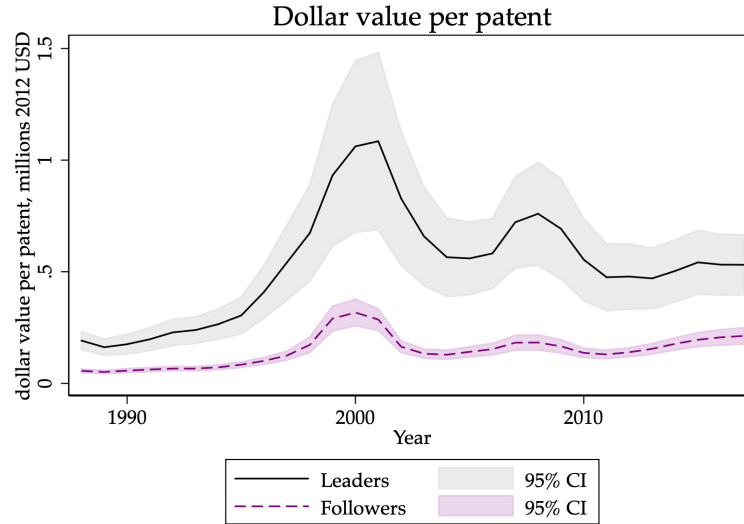


Figure A.4: Average dollar value per patent in USD 2012 millions using the estimates of [Kogan et al. \(2017\)](#). Leader indicates sales leaders in 4-digit SIC industries in a given year and followers are all other public firms in the sample. Three year moving averages. See Appendix A.1 for more details.

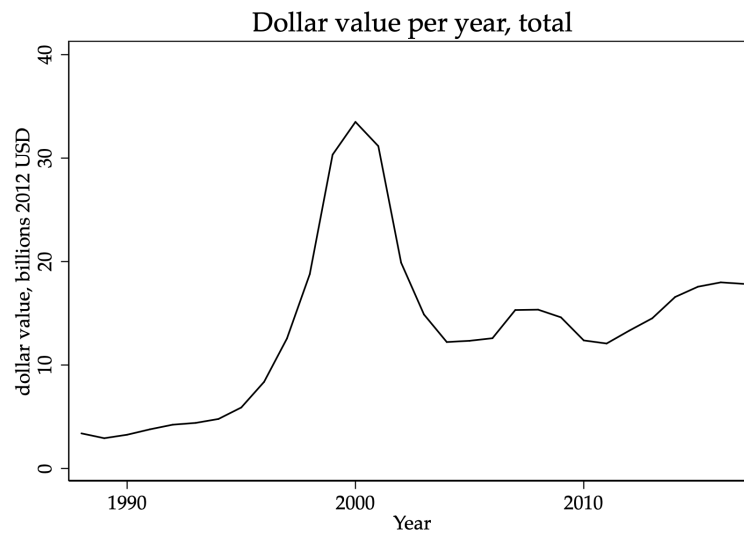


Figure A.5: Total value of patents generated each year summing [Kogan et al. \(2017\)](#)'s estimates across all firms. Three year moving average. See Appendix A.1 for more details.

	1994-2003	2004-2017	Change
Pat. val per R&D, leaders	10.08	9.76	-3.18
Pat. val per R&D, followers	3.98	3.05	-23.38
Ann. pat. growth, leaders	15.59	13.73	-11.95
Ann. pat. growth, followers	10.65	9.98	-6.31
Dollar value per patent, leaders	0.67	0.57	-14.91
Dollar value per patent, followers	0.18	0.16	-7.56

Table A.3: Alternative patent quality measures. Dollar values are in millions of 2012 USD. Change is percent change. See Appendix A.2 for more details.

A.4 TFP and Markup Estimation

Compustat data on U.S. public firms with the restrictions described in Appendix A.1 from 1962-2018 is used to estimate revenue-based total factor productivity (TFPR) and markups at the firm level. Firms without an industry classification, firm-years where acquisitions are more than 5% of assets ($ACQ/AT > 0.05$), and firm-years with non-positive employment ($EMP \leq 0$) or negative sales ($SALE \leq 0$) are also excluded. The sample includes around 3,000 firms per year, though this number varies over time.

Each firm’s capital stock $K_{i,t}$ is constructed by initializing the capital stock as PPEGT (total gross property, plant, and equipment) for the first year the firm appears. I then construct $K_{i,t+1}$ recursively:

$$K_{i,t+1} = K_{i,t} + I_{i,t+1} - \delta K_{i,t}$$

where PPENT (total net property, plant, and equipment) is used to capture the last two terms (net investment). I deflate the nominal capital stock using the Bureau of Economic Analysis (BEA) deflator for non-residential fixed investment.

In de Loecker and Warzynski (2012) the authors show that under a variety of pricing models firm i ’s markup at time t , μ_{it} , can be computed as a function of the output elasticity θ_{it}^V of any variable input and the variable input’s cost share of

revenue²⁸ :

$$\mu_{it} = \theta_{it}^V \frac{P_{it} Q_{it}}{P_t^V V_{it}} \quad (16)$$

where P_{it} is the output price of firm i 's good at time t , Q_{it} its output, P_t^V the price of the variable input and V_{it} the amount of the input used.

Following [de Loecker, Eeckhout, and Unger \(2020\)](#) I use COGS (cost of goods sold) deflated by the BEA's GDP deflator series as the real variable input cost $M_{i,t}$ of the firm. While the number of employees is well measured in Compustat and would be sufficient to estimate productivity, the wage bill is usually not available and would be needed to compute the labor cost share needed to compute the markup simultaneously with productivity.

For the results presented in this paper, I assume a Cobb-Douglas production function²⁹ for firm i in 2-digit SIC sector s in year t so that factor shares may vary across sectors but not over time:

$$Y_{i,s,t} = A_{i,s,t} M_{i,s,t}^{\beta_{M,s}} K_{i,s,t}^{\beta_{K,s}}$$

I use the variable SALE to measure firm output $Y_{i,s,t}$. I deflate SALE using the GDP deflator series to obtain real revenue at the firm level. I include firm and time fixed effects and obtain revenue-based TFP in logs (lower case variables denote variables in logs) by computing the residual (including fixed effects) of the following regressions for each 2-digit sector:

$$y_{i,t} = \alpha + \eta_t + \delta_i + \beta_{M,s} m_{i,t} + \beta_{K,s} k_{i,t-1} + \varepsilon_{i,t}.$$

In the above equation, $\beta_{M,s}$ captures the sector specific variable output elasticity, so I use equation 16 to obtain the markup from the estimated $\hat{\beta}_{M,s}$ and the inverse cost share $\frac{SALE}{COGS}$.

²⁸This approach requires several assumptions. First, the production technology must be continuous and twice differentiable in its arguments. Second, firms must minimize costs. Third, prices are set period by period. Fourth, the variable input has no adjustment costs. No particular form of competition among firms need be assumed.

²⁹Alternative estimation of a translog production function yielded similar estimates.

A.5 Industry Profit Shares

The competitive fringe assumption generates empirically plausible predictions about profit shares: the largest U.S. public firms (by sales) capture by far the largest share of industry profits (see Figure A.6).³⁰

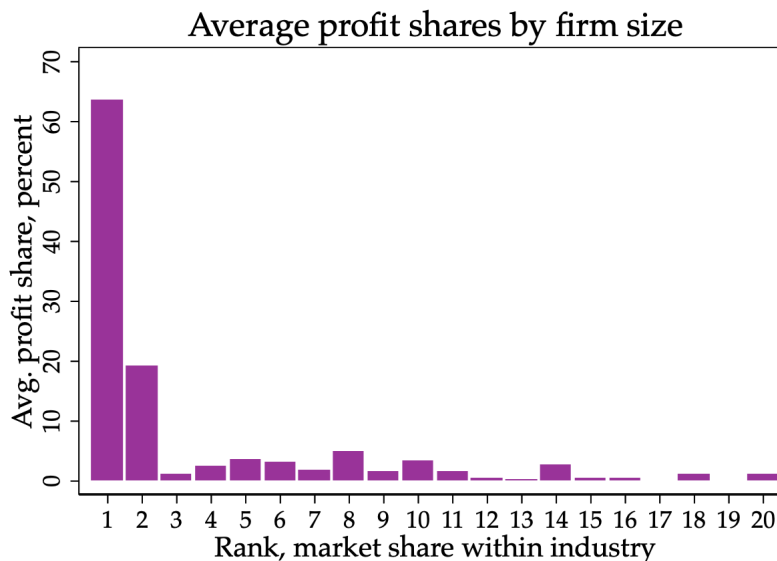


Figure A.6: Firms are ranked by market share (sales) within 4-digit SIC industries for a given year, and these ranks are compared to profit shares (firm’s own operating income (OIDBP) as a share of industry-total operating income). Averages across 4-digit sectors and across time for 1980-2017. See Appendix A.1 for more details.

A.6 Additional Model Validation Figures

Table A.4 shows non-targeted moments for the two subperiods, and columns three and four show the same moments between the two estimated model steady states. The last two columns show the model steady states where only the idea quality (IQ) or knowledge diffusion (KD) parameters change. Most of the changes in these non-targeted moments are due to the change in the idea quality parameters. Figure A.7

³⁰Following the estimation strategy of [de Loecker, Eeckhout, and Unger \(2020\)](#) to estimate TFP and markups at the firm level, TFP and sales share are correlated, and the figure looks similar if one uses a productivity ranking instead of sales-share based ranks.

	D:90s	D:00s	M:90s	M:00s	IQ	KD
Sale-weighted markup, %	27.47	33.21	32.58	34.92	35.11	32.60
Productivity gap, log points	63.24	67.33	29.52	41.12	39.93	29.58
Std. dev. TFPR growth, followers w/ pat., %	14.50	13.38	9.80	6.42	6.22	9.82
Leader share of R&D, %	41.32	47.04	30.47	57.69	56.87	30.45

Table A.4: Fit for changes in non-targeted moments. IQ is steady state where idea quality parameters λ, ϕ change to the 2000s estimated values in Table 5 and others are fixed. KD is steady state where the exit/knowledge diffusion shock δ_e varies to its 2000s value and other parameters are fixed. See Appendix A.2 for more details on moment construction.

plots the leaders' share of patents and R&D expenditure over time.

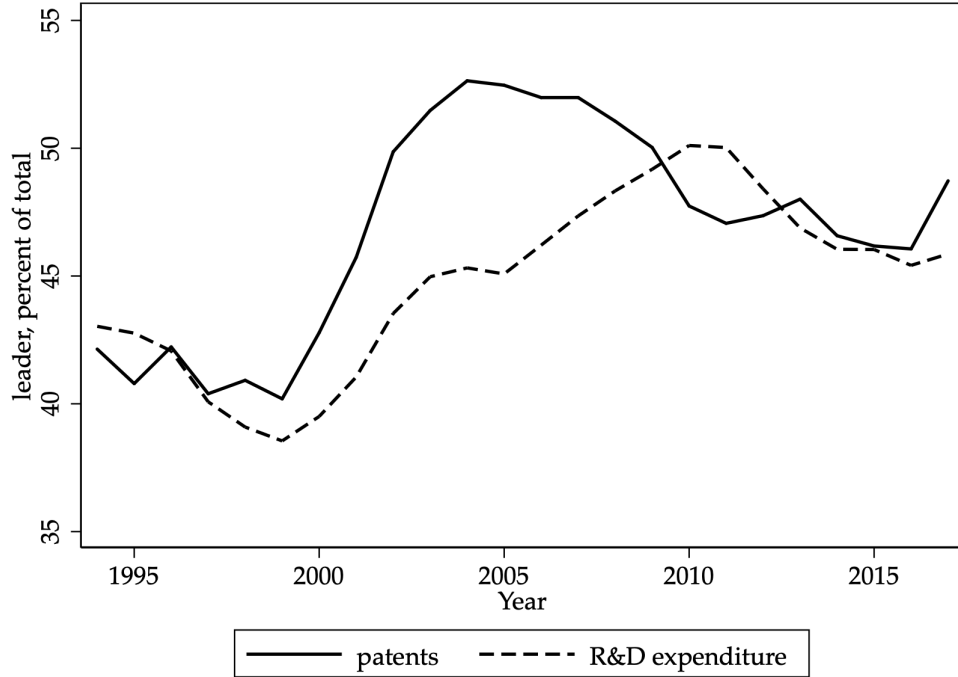


Figure A.7: Industry sales leader's share of industry R&D expenditure and patents, 3 year moving average, sale-weighted average across 4-digit SIC industries. See Appendix A.1 for more details.

A.6.1 Distinction from Akcigit and Ates (2023)

A key data moment that distinguishes this paper’s proposed explanation for changes in business dynamism from the declining knowledge diffusion hypothesis is the dispersion in followers’ TFPR growth rates conditional on patenting. Like this paper, Akcigit and Ates (2023) show that declining knowledge diffusion from leader to follower in their model raises in concentration, productivity gaps, markups, and concentration of R&D expenditures and patenting. This generates a decline in the overall dispersion of TFPR growth rates. However, declining knowledge diffusion does not change the idea quality distribution firms face, which is either a one step λ improvement, or quick catch up by the follower to the leader which occurs with probability ϕ when firms patent. Since productivity gaps get larger, their theory predicts that conditional on innovating these catch up events will involve larger productivity improvements when the rate of knowledge diffusion is low.

The data in Table A.4 show the opposite: followers’ TFPR growth rates conditional on patenting became less dispersed over time, which is consistent with declining dispersion in the quality of new ideas. This distinction is important for understanding the effects of size-dependent R&D subsidies, which would yield different output in terms of productivity growth in the two models (with lower idea quality, generating the same amount of growth becomes more costly). The Related Literature section in Section 1 discusses other differences between the two hypotheses, such as the ability to explain the productivity slowdown.

A.6.2 Sector-Level Concentration and Productivity Slowdowns

An empirical exploration of the causal relationships among productivity growth and concentration is beyond the scope of this paper. However, given the sectoral heterogeneity in the decline in laggards patent quality in Figure A.3 suggesting that this phenomena differs across industries, we might expect rising concentration and the productivity slowdown to be correlated at the sector level. I use data from Bureau of Economic Analysis estimates of multifactor productivity³¹ at the 3-digit NAICS level and Compustat to check the association between the change in the leader’s market share and the change in the sector’s average productivity growth rate from 1994-2003

³¹<https://www.bea.gov/data/special-topics/integrated-industry-level-production-account-klems>

to 2004-2017. Sectors with greater slowdowns in productivity growth also saw greater increases in concentration on average (Figure A.8).

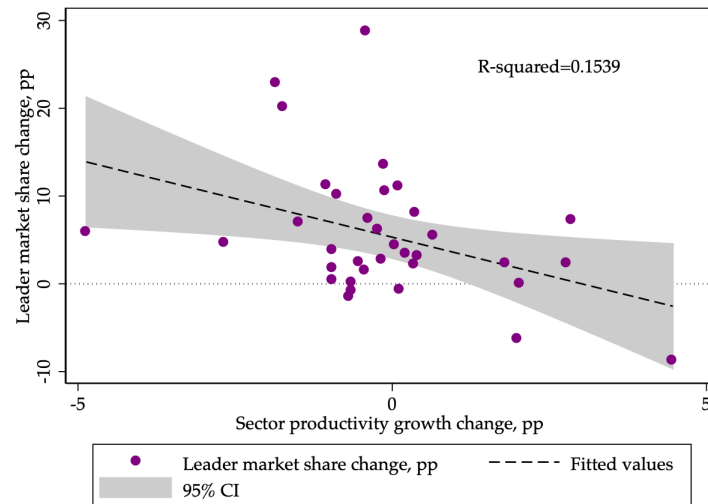


Figure A.8: Each dot is a 3-digit NAICS sector. Dots show the change in average annual total factor productivity growth from the BEA Integrated Industry-Level Production Accounts between 1994-2003 and 2004-2017 (x-axis) plotted against the change in the largest firm's average share of industry sales between 1994-2003 and 2004-2017 in Compustat (y-axis). Excludes oil, gas, and mining industries (211, 212, 213).

B Online Model Appendix

B.1 Proof Prices Depend on Relative Quality

Below I show that the firms' pricing strategies depend only on their relative quality, not the level of their own or their rival's quality. Let firm i be the industry follower so that the fringe's quality $\frac{q_f}{q_i} = \zeta$ and $\frac{q_{-i}}{q_i} = \lambda^{-m_i}$ where $m_i \leq 0$. Plugging the final good firm's demand for good i into the definition of the market share and using the definition of the price index yields:

$$\begin{aligned} s_i &= q_i^{\epsilon-1} \left(\frac{p_i}{P_j} \right)^{1-\epsilon} \\ &= \frac{q_i^{\epsilon-1} p_i^{1-\epsilon}}{q_i^{\epsilon-1} p_i^{1-\epsilon} + q_{-i}^{\epsilon-1} p_{-i}^{1-\epsilon} + q_f^{\epsilon-1} \eta^{1-\epsilon}} \\ &= \frac{1}{1 + \left(\frac{q_{-i}}{q_i} \right)^{\epsilon-1} \left(\frac{p_{-i}}{p_i} \right)^{\epsilon-1} + \left(\frac{q_f}{q_i} \right)^{\epsilon-1} \left(\frac{p_i}{\eta} \right)^{\epsilon-1}}. \end{aligned}$$

Now using the large firms' pricing policies:

$$s_i = \left(1 + (\lambda^{-m_i})^{\epsilon-1} \left(\frac{\frac{\epsilon - (\epsilon - \frac{1}{\beta})s_i}{\epsilon - (\epsilon - \frac{1}{\beta})s_i - 1}}{\frac{\epsilon - (\epsilon - \frac{1}{\beta})s_{-i}}{\epsilon - (\epsilon - \frac{1}{\beta})s_{-i} - 1}} \right)^{\epsilon-1} + \zeta^{\epsilon-1} \left(\frac{\epsilon - (\epsilon - \frac{1}{\beta})s_i}{\epsilon - (\epsilon - \frac{1}{\beta})s_i - 1} \right)^{\epsilon-1} \right)^{-1}.$$

A similar argument shows that s_{-i} is independent of quality levels. Thus there is a mapping from technology gaps to market shares and prices that is independent of quality levels. ■

B.2 Value Function Boundary Equations

For a firm that's the furthest possible number of steps behind its competitor (that is, at gap $-\bar{m}$) and absolute quality level q_{ijt} :

$$\begin{aligned} r_t V_{-\bar{m},t}(q_{ijt}) - \dot{V}_{-\bar{m},t}(q_{ijt}) = & \max_{x_{-\bar{m},t}} \left\{ \pi(-\bar{m}, q_{ijt}) - \alpha \frac{(x_{-\bar{m},t})^\gamma}{\gamma} q_{ijt}^{\frac{1}{\beta}-1} \right. \\ & + x_{-\bar{m},t} \sum_{n_t=-\bar{m}+1}^{\bar{m}} \mathbb{F}_m(n_t) [V_{nt}(\lambda^{n_t-(-\bar{m})} q_{ijt}) - V_{-\bar{m},t}(q_{ijt})] \\ & + x_{\bar{m},t} (V_{-\bar{m},t}(\lambda q_{ijt}) - V_{-\bar{m},t}(q_{ijt})) \\ & \left. + \delta_e (0 - V_{-\bar{m},t}(q_{ijt})) \right\}. \end{aligned}$$

The difference between this and equation 6 is in the third line, where if the firm's competitor innovates, there is a spillover that causes the firm at gap $-\bar{m}$ to improve its quality by λ .

For a firm at gap $\bar{m}$ the value function is:

$$\begin{aligned} r_t V_{\bar{m},t}(q_{ijt}) - \dot{V}_{\bar{m},t}(q_{ijt}) = & \max_{x_{\bar{m},t}} \left\{ \pi(\bar{m}, q_{ijt}) - \alpha \frac{(x_{\bar{m},t})^\gamma}{\gamma} q_{ijt}^{\frac{1}{\beta}-1} \right. \\ & + x_{\bar{m},t} (V_{\bar{m},t}(\lambda q_{ijt}) - V_{\bar{m},t}(q_{ijt})) \\ & + x_{-\bar{m},t} \sum_{n_t=-\bar{m}+1}^{\bar{m}} \mathbb{F}_{-\bar{m}}(n_t) [V_{nt}(q_{ijt}) - V_{\bar{m},t}(q_{ijt})] \\ & \left. + \delta_e (0 - V_{\bar{m},t}(q_{ijt})) \right\}. \end{aligned}$$

B.3 Derivation of Final Output

The section derives the expression for final output in equations 9 and 10. Dropping the time subscript t , plugging the pricing strategies in equation 5 into the demand curve (4) to obtain the output of each good and plugging these outputs into equation

(3) and equation (3) into equation (2) simplifies as:

$$\begin{aligned}
Y &= \frac{1}{1-\beta} \left(\int_0^1 K_j^{1-\beta} dj \right) L^\beta \\
&= \frac{1}{1-\beta} \left(\int_0^1 \left(\sum_{i=1}^3 q_i^{\frac{\epsilon-1}{\epsilon}} (q_i^{\epsilon-1} \left(\frac{p_i}{P_j} \right)^{-\epsilon} \left(\frac{P_j}{P} \right)^{-\frac{1}{\beta}} L)^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1} 1-\beta} dj \right) L^\beta \\
&= \frac{L}{1-\beta} P^{\frac{1-\beta}{\beta}} \left(\int_0^1 P_j^{\epsilon(1-\beta)-\frac{1-\beta}{\beta}} \left(\sum_{i=1}^3 q_i^{\epsilon-1} p_i^{1-\epsilon} \right)^{\frac{\epsilon(1-\beta)}{\epsilon-1}} dj \right) \\
&= \frac{L}{1-\beta} P^{\frac{1-\beta}{\beta}} \left(\int_0^1 P_j^{-\frac{1-\beta}{\beta}} dj \right).
\end{aligned}$$

The demand shifter $P^{\frac{1}{\beta}} L$ index is common to all firms and can therefore be factored out of the integral (and normalized to one since I assume that there is no population growth). The quality-adjusted price index P_j of each sector falls as the qualities of the products in the sector grow, and the exponent is negative for all $\beta \in (0, 1)$ so Y is increasing in firms' qualities.

Common to all firms with a particular technology gap m are the prices p_m of the firm at gap m and its competitor at $-m$, p_{-m} . At time t , therefore, Y can be expressed as:

$$Y_t = \frac{1}{2} \frac{L}{1-\beta} P^{\frac{1-\beta}{\beta}} \sum_{m=-\bar{m}}^{\bar{m}} Q_{mt},$$

where Q_{mt} is defined as:

$$Q_{m,t} = \begin{cases} (p_m^{1-\epsilon} + (\lambda^{-m})^{\epsilon-1} p_{-m}^{1-\epsilon} + \zeta^{\epsilon-1} \eta^{1-\epsilon})^{\frac{1-\beta}{\beta(\epsilon-1)}} \int_0^1 q_{i,t}^{\frac{1-\beta}{\beta}} \mathbb{1}_{\{i \in \mu_{mt}\}} di & \text{for } m \in \{-\bar{m}, \dots, 0\} \\ (p_m^{1-\epsilon} + (\lambda^{-m})^{\epsilon-1} p_{-m}^{1-\epsilon} + (\zeta \lambda^{-m})^{\epsilon-1} \eta^{1-\epsilon})^{\frac{1-\beta}{\beta(\epsilon-1)}} \int_0^1 q_{i,t}^{\frac{1-\beta}{\beta}} \mathbb{1}_{\{i \in \mu_{mt}\}} di & \text{for } m \in \{1, \dots, \bar{m}\} \end{cases}.$$

B.4 Output Growth on Balanced Growth Path

To understand how aggregate output evolves, this section studies the evolution of $\tilde{Q}_{m,t}$ (defined in equation 14) between t and $t + dt$ for all m . These expressions are similar to those for the stationary distribution (equations 11-13) because they are based on the movement of firms to different technology gaps from their rival, but account for the quality improvements that occur because of innovation.

Assuming fixed distribution $\mu_{mt} = \mu_m$ for all m, t :

$$\dot{Q}_{mt} = \int_0^1 q_{m,t+dt,i}^{\frac{1-\beta}{\beta}} \mathbb{1}_{\{i \in \mu_m\}} di - \int_0^1 q_{m,t,i}^{\frac{1-\beta}{\beta}} \mathbb{1}_{\{i \in \mu_m\}} di.$$

that is, quality growth at gap m is due to the change in an index of the qualities of all the firms with technology gap m . Consider an arbitrary $m \in (-\bar{m}, \bar{m})$ ($-\bar{m}$ and $\bar{m}$ are special cases because of spillovers). A portion of firms at m at t innovate to a different gap, and another portion leave gap m because their competitor innovates. Because all firms at gap m choose the same arrival rate x_m , these are a random sample of the firms at gap m at time t . The outflows from $\dot{Q}_m$ are:

$$-(x_m + x_{-m}) \int_0^1 q_{m,t,i}^{\frac{1-\beta}{\beta}} \mathbb{1}_{\{i \in \mu_m\}} di = -(x_m + x_{-m}) \tilde{Q}_m.$$

The inflows to m 's quality index come from two sources. First, some firms innovate into position m from a lower position n , improving their quality by λ^{m-n} . The probability they innovate and reach gap m is given by $x_n F_n(m)$. Some firms fall back to m from a higher gap n because their competitor innovates to $-m$. The probability their competitor reaches $-m$ is given by $x_{-n} F_{-n}(-m)$. So cumulative inflows are:

$$\sum_{n=-\bar{m}}^{m-1} x_n F_n(m) (\lambda^{(m-n)})^{\frac{1-\beta}{\beta}} \tilde{Q}_n + \sum_{n=m+1}^{\bar{m}} x_{-n} F_{-n}(-m) \tilde{Q}_n.$$

Putting it together:

$$\dot{Q}_{mt} = \sum_{n=-\bar{m}}^{m-1} x_n F_n(m) (\lambda^{(m-n)})^{\frac{1-\beta}{\beta}} \tilde{Q}_n + \sum_{n=m+1}^{\bar{m}} x_{-n} F_{-n}(-m) \tilde{Q}_n - (x_m + x_{-m}) \tilde{Q}_m. \quad (17)$$

At the lowest gap there are spillovers when the competitor innovates:

$$\dot{Q}_{-\bar{m}t} = \sum_{n=-\bar{m}+1}^{\bar{m}} x_{-n} F_{-n}(\bar{m}) \tilde{Q}_n + x_{\bar{m}} (\lambda^{\frac{1-\beta}{\beta}} - 1) \tilde{Q}_{-\bar{m}} - x_{-\bar{m}} \tilde{Q}_{-\bar{m}}. \quad (18)$$

At the highest gap the leading firm does not exit that gap when they innovate:

$$\dot{Q}_{\bar{m}t} = \sum_{n=-\bar{m}}^{\bar{m}-1} x_n F_n(\bar{m}) (\lambda^{(m-n)})^{\frac{1-\beta}{\beta}} \tilde{Q}_n + x_{\bar{m}} (\lambda^{\frac{1-\beta}{\beta}} - 1) \tilde{Q}_{\bar{m}} - x_{-\bar{m}} \tilde{Q}_{\bar{m}}. \quad (19)$$

Given equations 17, 18, and 19, on a balanced growth path where $\frac{\dot{Q}_{mt}}{Y_t}$ is constant, it's sufficient to assume $\frac{\tilde{Q}_{mt}}{Y_t}$ is constant over time for all $m \in [-\bar{m}, \bar{m}]$. Differentiating $\frac{\tilde{Q}_{mt}}{Y_t}$ with respect to time yields:

$$\begin{aligned}\left(\frac{\dot{\tilde{Q}}_m}{Y}\right) &= \frac{\dot{\tilde{Q}}_m}{Y} - \frac{\tilde{Q}_m}{Y} \frac{\dot{Y}}{Y} \\ &= \frac{\dot{\tilde{Q}}_m}{Y} - g \frac{\tilde{Q}_m}{Y}.\end{aligned}$$

Imposing that the left hand side is zero implies:

$$\frac{\dot{\tilde{Q}}_m}{Y} = g \frac{\tilde{Q}_m}{Y}.$$

The vector on the left hand side is defined above by the flow equations (17), (18), and (19) divided by output. Use those equations to form a matrix A that captures the flow equations:

$$\frac{\dot{\tilde{Q}}_m}{Y} = A \frac{\tilde{Q}_m}{Y} = g \frac{\tilde{Q}_m}{Y}.$$

The values in A depend on λ, ϕ , and x_m . The above equation means that the growth rate g is an eigenvalue of the matrix A and $\frac{\tilde{Q}_m}{Y}$ is the corresponding eigenvector of A. If there is only one positive, real eigenvalue of A then there exists only one such balanced growth path where the contribution of the growth of the quality index of each technology gap to the total growth rate is constant and the growth rate of the economy is constant.

B.5 Growth Decomposition Formulas

Formally, the growth decompositions in Section 5.1 are computed as follows. Define a function capturing the model's steady state growth rate $g(x, \lambda, \phi, \boldsymbol{\theta})$ where x is the innovation policy, λ is the minimum quality improvement, ϕ is the step size distribution parameter, and $\boldsymbol{\theta}$ is a vector of the other model parameters which are held at 1990s values throughout the growth decomposition exercise and is thus suppressed in the formulas that follow for ease of exposition. Further define the innovation policy in steady state as a function of parameters $x(\lambda, \phi, \boldsymbol{\theta})$. The decompositions are:

$$\begin{aligned}\text{strategic effect} &= \frac{g(x(\lambda_{1990s}, \phi_{1990s}), \lambda_{1990s}, \phi_{1990s}) - g(x(\lambda_{2000s}, \phi_{2000s}), \lambda_{1990s}, \phi_{1990s})}{g(x(\lambda_{1990s}, \phi_{1990s}), \lambda_{1990s}, \phi_{1990s}) - g(x(\lambda_{2000s}, \phi_{2000s}), \lambda_{2000s}, \phi_{2000s})}, \\ \text{first order effect} &= \frac{g(x(\lambda_{1990s}, \phi_{1990s}), \lambda_{1990s}, \phi_{1990s}) - g(x(\lambda_{1990s}, \phi_{1990s}), \lambda_{2000s}, \phi_{2000s})}{g(x(\lambda_{1990s}, \phi_{1990s}), \lambda_{1990s}, \phi_{1990s}) - g(x(\lambda_{2000s}, \phi_{2000s}), \lambda_{2000s}, \phi_{2000s})}.\end{aligned}$$

B.6 Alternate Model With No Competitive Fringe

This appendix solves the full dynamic model without the competitive fringe, assuming there are just the two incumbent varieties in each industry. There is still the possibility of exogenous entry/exit, though this assumption can be relaxed as well. In this setup the quality leader always has at least 50% market share, unlike in the data. This assumption also gives empirically counterfactual predictions that the profit shares of total industry profits of the market leader and the other firm in the industry are relatively similar, contradicting the pattern shown in Figure A.6.

Nonetheless, the main results carry through under this alternate assumption. The algorithm for finding the steady state remains the same, plugging in the pricing functions of the firms, illustrated in Figure B.1.

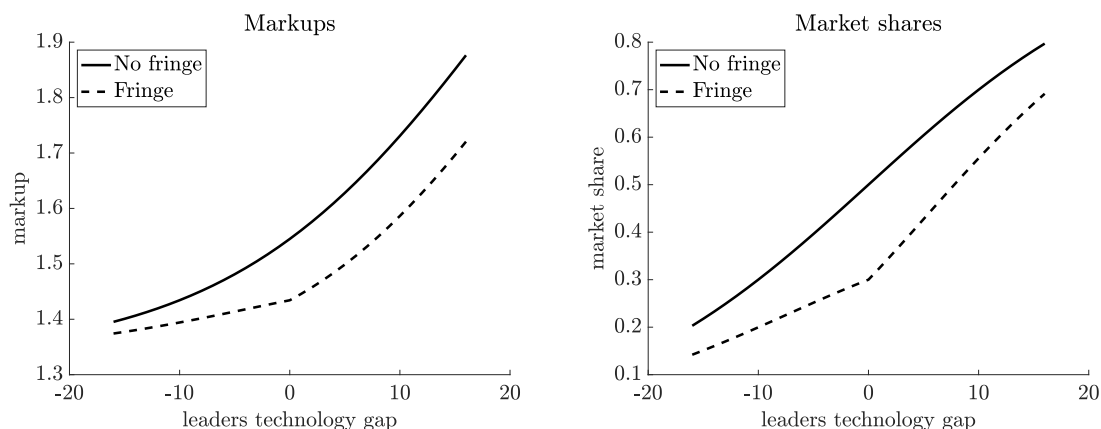


Figure B.1: Leader's markup (price over marginal cost) and market share as a function of the leader's technology gap m in the model using the 1990s calibration in Table 1 in the version of the model with no competitive fringe compared with the baseline model.

Idea Quality. Table B.1 gives the results of the declining idea quality experiment from section 5.1 using the same parameters as in Table 1 for the model with no fringe and Figure B.2 shows the policy functions and stationary distributions. As before, changing ϕ has a level effect on total innovation effort but also changes the location of R&D from laggard firms to leading firms.

The level of the growth rates and the change in the growth rate from one steady state to the other due to declining idea quality are very similar to the baseline model

<u>Moment</u>	<u>Data</u>			<u>No Fringe Model</u>		
	1990s	2000s	Chg. (pp)	1990s	2000s λ, ϕ	Chg. (pp)
TFP growth	1.6	0.7	-0.9	1.9	0.7	-1.2
Leader market share	43.2	48.1	4.9	59.8	61.9	2.1
R&D/GDP	1.8	1.9	0.1	3.0	1.5	-1.5
Profits/GDP	7.7	9.2	1.5	11.6	13.3	1.7
Leadership turnover	13.7	9.3	-4.5	15.3	11.5	-3.8
Pat. qual., followers	19.0	11.9	-7.2	17.6	10.7	-7.0
Pat. qual., leaders	6.9	4.9	-2.0	8.2	7.1	-1.1
Patent quality, std. dev., %	25.8	19.5	-6.3	28.2	20.7	-7.6

Table B.1: “Data” columns show the levels and changes in the key moments in the data. Model columns show the levels and changes of the moments in the equilibrium of the model without the competitive fringe under the parameterization in Table 1 (“1990s”) and the counterfactual equilibrium with the same parameterization except varying λ and ϕ to their estimated value in the 2000s from Table 5 (“2000s λ, ϕ ”).

with the fringe. The increase in concentration is smaller, about 43% of the increase in the data instead of 89%. As for the growth decomposition, the effects of the firms’ innovation responses is slightly smaller (44.5%), and the first order effect of lowering the probability of radical innovations is a bit larger than in the baseline model with the fringe (86.1%).

Superstar Firms. The second robustness considers an increase in the elasticity of substitution in the model with no fringe (Table B.2) to match a 4.9pp rise in concentration. Like in the baseline model this results in a small productivity slowdown and a rise in concentration, but an increase in patent quality and a decline in the profit share, so is not consistent with the data.

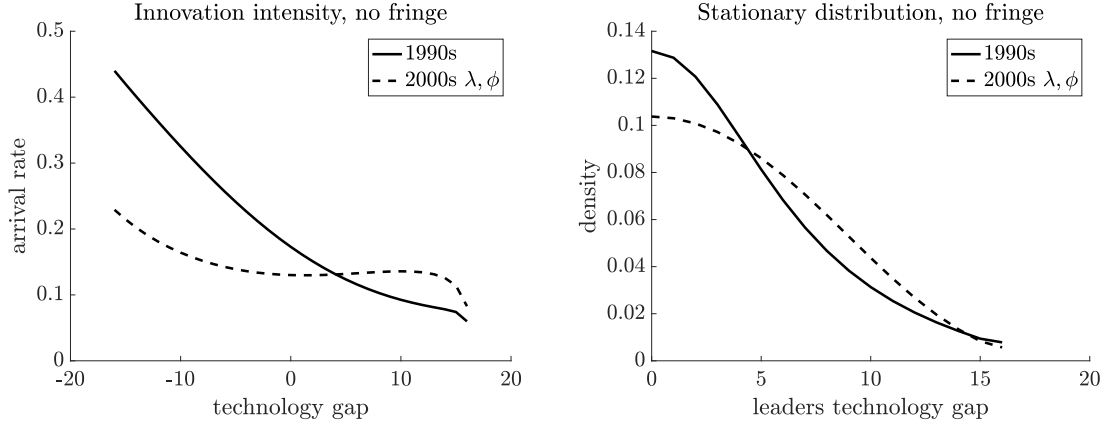


Figure B.2: Innovation policies x as a function of the technology gap for all firms (panel a) and the stationary distribution over the technology gap of the leader (panel b) using the 1990s calibration in Table 1 in the version of the model with no competitive fringe.

<u>Moment</u>	<u>Data</u>			<u>No Fringe Model</u>		
	1990s	2000s	Chg. (pp)	1990s	$\epsilon = 6.57$	Chg. (pp)
TFP growth	1.6	0.7	-0.9	1.9	1.7	-0.2
Leader market share	43.2	48.1	4.9	59.8	64.7	4.9
R&D/GDP	1.8	1.9	0.1	3.0	2.3	-0.8
Profits/GDP	7.7	9.2	1.5	11.6	8.9	-2.8
Leadership turnover	13.7	9.3	-4.5	15.3	14.5	-0.8
Pat. qual., followers	19.0	11.9	-7.2	17.6	26.5	8.9
Pat. qual., leaders	6.9	4.9	-2.0	8.2	11.7	3.4
Patent quality, std. dev., %	25.8	19.5	-6.3	28.2	45.4	17.1

Table B.2: “Data” columns show the levels and changes in the key moments in the data. Model columns show the levels and changes of the moments in the equilibrium of the model without the competitive fringe under the parameterization in Table 1 (“1990s”) and the counterfactual equilibrium with the same parameterization except varying ϵ .

B.7 Robustness: Deterministic Step Sizes

The results do not depend crucially on randomness in the quality of new ideas for a given firm type m . What matters for the firm value functions and policies is the expected change in the value function conditional on innovating. This section does the following:

1. Take the 1990s calibration ($\bar{m} = 16, \lambda = 1.048, \phi = 0.99$) and compute the expected number of steps a firm with gap $m \in \{-16, -15, \dots, 14, 15, 16\}$ makes using the baseline step size distribution $\mathbb{F}_m(n)$.
2. This gives an expected step size for each type m taking values

$$\{9.8959, 9.0577, \dots, 1.0081, 1, 1\}.$$

3. Round these back on to an integer grid to keep quality gaps on the grid so that the deterministic step sizes take the values:

$$\{10, 9, \dots, 1, 1, 1\}.$$

4. Solve the model under this assumption on the distribution $\tilde{\mathbb{F}}_m(n)$ which is now degenerate on a particular step size n for each type m . The rest of the solution method should be exactly the same as the baseline model.
5. Repeat the above exercise under the patent quality parameters for the 2000s and resolve the model. The results are in Table B.3. Qualitatively a decline in patent quality affects the growth rate, concentration, profit share, etc. in the same way as in the baseline. The levels are not identical because of the use of a discrete grid and the rounding in step (c).

	M:1990s	M:2000s λ, ϕ
TFP growth, %	1.1	0.3
Avg. leader market share, %	36.8	37.8
R&D share of GDP, %	1.2	0.5
Profit share of GDP, %	6.7	7.7
Leadership turnover, %	17.4	10.8
Patent quality, avg., followers, %	12.4	6.1
Patent quality, avg., leaders, %	9.1	6.3
Patent quality, std. dev., %	2.1	1.3

Table B.3: Model fit for targeted moments, deterministic step sizes. See text for details.

B.8 Alternative Calibration of Idea Quality

This section presents an alternative calibration for the idea quality parameters to exactly match the percent declines in patent quality for leaders and followers. The results are in Table B.4. The estimated values of $\lambda = 1.0383$ and $\phi = 1.24$. These values decrease rather than increase concentration, but the effect on productivity growth is robust. The joint estimation favors larger changes in ϕ to match a simultaneous rise in concentration and profits.

The same growth decompositions from section B.5 imply that the strategic effect accounts for 52% of the change in the growth rate under these parameter changes and the first order effect is 72%, so the role of the strategic effect is between 28-52%, similar to the baseline model.

B.9 Cutting Step Size in Half

Table B.5 shows the results of an experiment cutting the step size λ in half but doubling the number of firm types $\bar{m}$ so that the furthest ahead leader has the same market share in the two economies. While this change reduces patent quality and the growth rate, it does not change the standard deviation of patent quality or concentration.

	<u>Data</u>	<u>Model</u>	
		$\lambda=1.04, \phi=1.24$	$\lambda=1.05, \phi=1.72$
TFP growth, %	-58.0	-62.8	-65.1
Avg. leader market share, %	11.3	-6.4	10.2
R&D share of GDP, %	5.2	-56.9	-51.3
Profit share of GDP, %	19.3	7.9	19.5
Leadership turnover, %	-32.5	-15.4	-24.3
Patent quality, avg., followers, %	-37.7	-37.7	-44.3
Patent quality, avg., leaders, %	-28.6	-28.6	-13.6
Patent quality, std. dev., %	-24.3	-33.3	-30.5

Table B.4: Percent changes in each moment, data, and from 1990s model steady state to steady state with idea quality parameters calibrated to match the decline in patent quality. Last column shows the effects of the baseline estimated changes in idea quality. See text for details.

B.10 Patent Quality and Innovation Policies

There is a tight connection between expected patent quality and incentives to innovate in the model. In the case where $\gamma = 2$ and defining expected patent quality of type m as:

$$\begin{aligned}
EPQ_m &\equiv \sum_{n=m+1}^{\bar{m}} \mathbb{F}_m(n) \left(\frac{(\lambda^{n_t-m})^{\frac{1}{\beta}-1} v_{nt} - v_{mt}}{v_{mt}} \right) \\
v_m EPQ_m &= \sum_{n=m+1}^{\bar{m}} \mathbb{F}_m(n) \left((\lambda^{n_t-m})^{\frac{1}{\beta}-1} v_{nt} - v_{mt} \right) \\
v_m EPQ_m &= x_m \alpha \\
x_m &= \frac{v_m EPQ_m}{\alpha}.
\end{aligned}$$

Figure B.3 shows the value function constants v_m under different values of market power ϵ . They are higher for all m when firms have more market power (ϵ is lower).

	D: 1990s	D: 2000s	M:1990s	M: $\frac{1}{2}\lambda_{1990s}$
TFP growth, %	1.57	0.66	1.58	1.01
Avg. leader market share, %	43.23	48.12	42.62	41.89
R&D share of GDP, %	1.80	1.89	1.76	1.21
Profit share of GDP, %	7.71	9.20	7.11	7.57
Leadership turnover, %	13.74	9.28	14.16	12.50
Patent quality, avg., followers, %	19.02	11.86	16.67	14.26
Patent quality, avg., leaders, %	6.87	4.91	8.44	6.38
Patent quality, std. dev., %	25.75	19.49	27.61	28.44

Table B.5: First two columns show targeted moments for 1990s and 2000s and third column shows model steady state for 1990s. Fourth column cuts the step size to half its 1990s value and doubles the maximal gap $\bar{m}$.

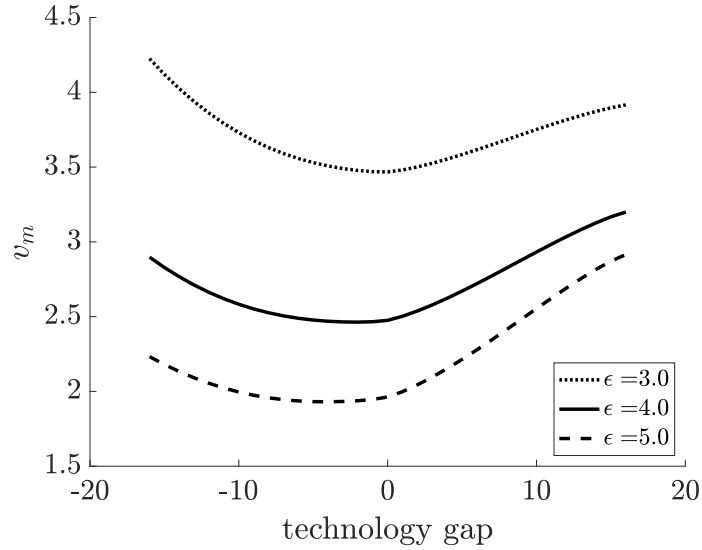


Figure B.3: Value function constants v_m under different values of the elasticity of substitution within sectors ϵ .

B.11 Extra Model Figures

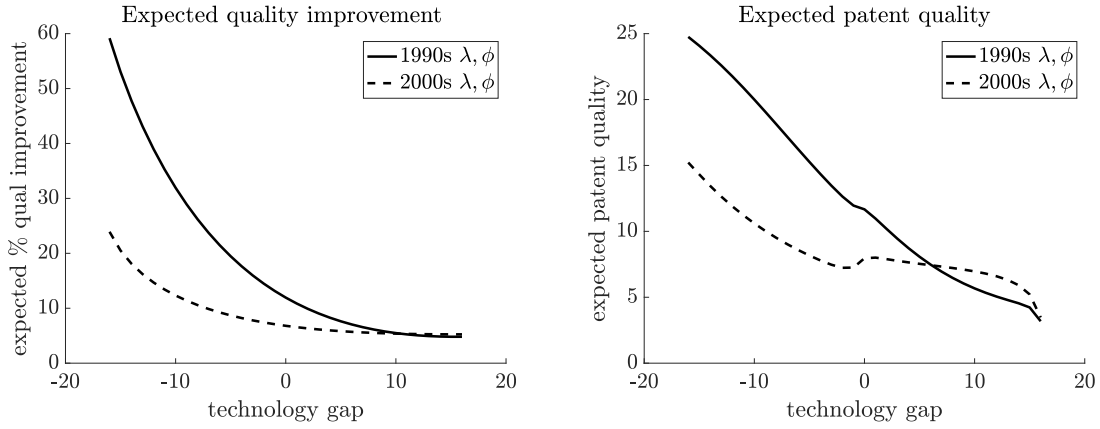


Figure B.4: Expected idea and patent quality as a function of the technology gap comparing the estimated values of λ, ϕ , holding other parameters fixed at 1990s values.

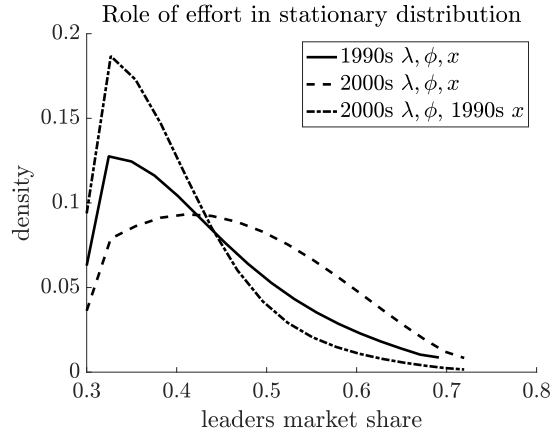


Figure B.5: Distribution of sectors over the leader's technology gap. Solid line is the 1990s steady state (Table 1). Dashed line is a counterfactual equilibrium where λ and ϕ change to their 2000s values and other parameters are fixed. Dash-dotted line is a counterfactual where the innovation policy function x is fixed at 1990s and innovation parameters λ, ϕ change to 2000s values.

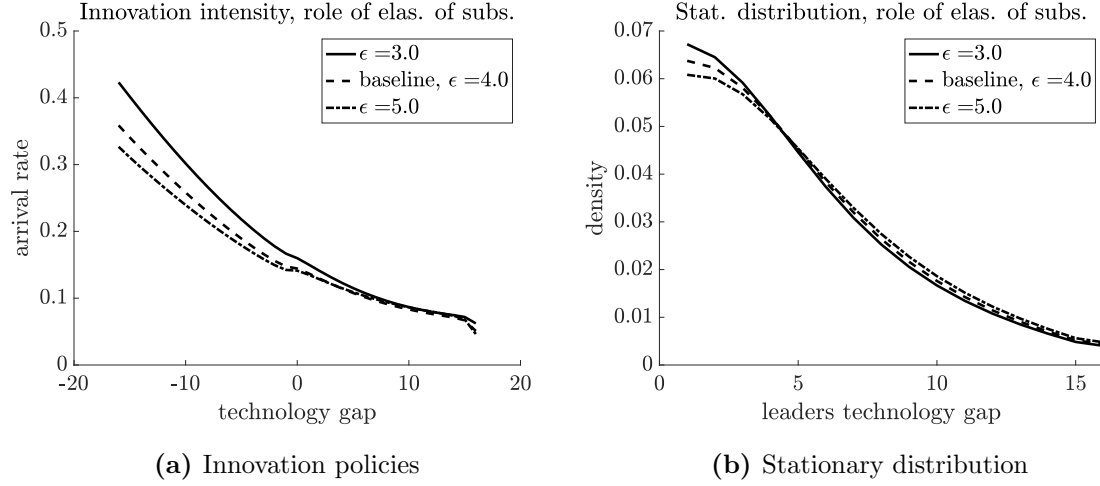


Figure B.6: Effect of the elasticity of substitution parameter ϵ on innovation policies x as a function of the technology gap for all firms (panel a) and the stationary distribution of sectors over the technology gap of the leader (panel b) under the calibration in Table 1. $\epsilon = 3$ is the market power experiment (dotted line), $\epsilon = 4$ is the baseline (solid line), and $\epsilon = 5$ is the superstar firm experiment (dashed line).

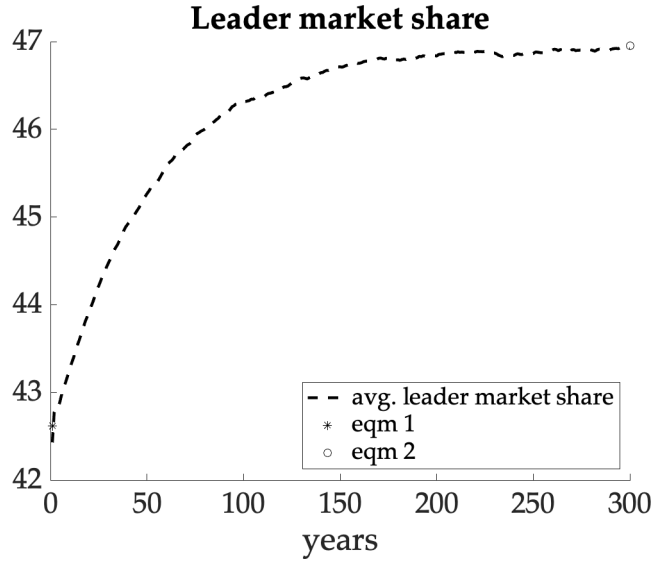


Figure B.7: Long run transition dynamics for concentration over the entire transition path to steady state with lower idea quality.

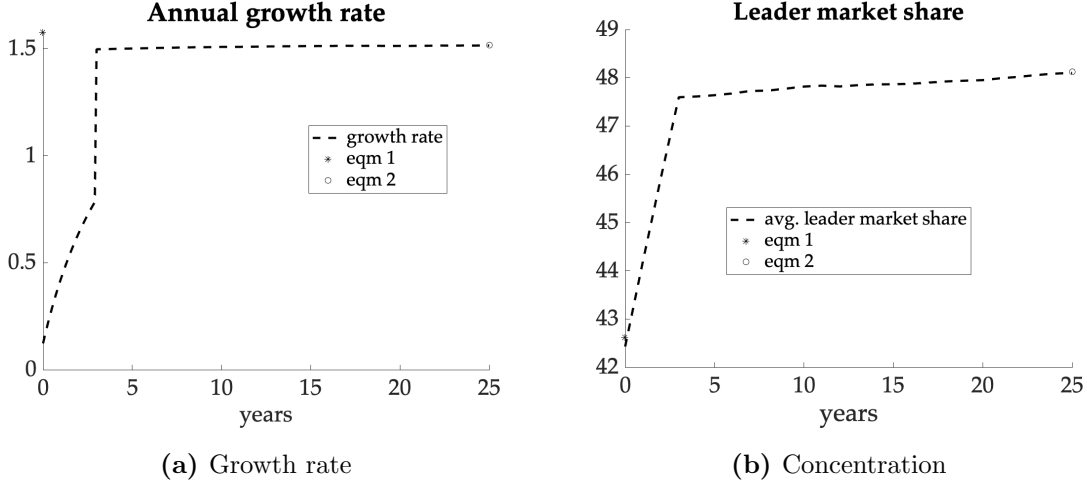


Figure B.8: Transition dynamics for productivity growth and concentration over the transition to higher competition steady state ($\epsilon = 5$). Details in the text and Appendix C.3.

C Online Numerical Appendix

C.1 Simulated Method of Moments Estimation

Let $M_j(\theta)$ denote the steady state value of moment j in the model as a function of the model parameters in vector θ . Let D_j denote the same moment in the data. The simulated method of moments estimation procedure seeks to find the vector of parameters θ^* that solves:

$$\min_{\theta} \sum_{j=1}^J w_j \left| \frac{M_j(\theta) - D_j}{\frac{1}{2}M_j(\theta) + \frac{1}{2}D_j} \right|$$

for J moments. I set the weights w_j such that the moments of interest, productivity growth and concentration, are weighted 5 times more than the other moments.

C.2 Solution Algorithm

For a given set of parameter values, the solution algorithm involves first guessing a steady state interest rate. Given this interest rate, solve the value functions for each technology gap by policy function iteration using the fact that $\dot{v}_{mt} = 0$ on a balanced growth path. This process yields the optimal innovation policies of firms at

each technology gap. Given the policy functions the stationary distribution of firms over technology gaps can be obtained by solving the system of equations described in section 3.6. To obtain the growth rate of output, solve the system described in Appendix B.4. Check whether this growth rate is consistent with the interest rate guess using the household's Euler equation: $r = g\psi + \rho$. Update the guess of the interest rate and repeat until the interest rate guess and the interest rate implied by the resulting growth rate and the Euler equation are consistent. To obtain micro-level moments, I simulate a discrete time version of the model with ten subperiods per year for a panel of 3000 firms for 400 years after the model reaches the steady state distribution over technology gaps.

C.3 Transition Dynamics

This appendix details the computational approach to solving the model's transition dynamics. The approach is similar to [Cavenaile, Roldan-Blanco, and Schmitz \(2022\)](#) but requires firm-level simulations at each iteration. The economy begins in the initial (1990s) steady state in period $t = 1$ and arrives at the new steady state (2000s) by T , where T is large:

1. Guess an interest rate path $\mathbf{r} = \{r_1, r_{1+dt}, r_{1+2dt}, \dots, r_T\}$.
2. Given the steady state values $v_{m,T}$ assumed at T , solve backward for innovation policies at $T - dt$ as:

$$x_{m,T-dt} = \begin{cases} \left(e^{-r_T dt} \frac{\sum_{n=m+1}^{\bar{m}} \mathbb{F}_{m,T-dt}(n) [(\lambda^{n-m})^{\frac{1}{\beta}-1} v_{n,T} - v_{m,T}]}{\alpha} \right)^{\frac{1}{\gamma-1}} & \text{for } m < \bar{m} \\ \left[e^{-r_T dt} \frac{1}{\alpha} (\lambda^{\frac{1}{\beta}-1} - 1) v_{\bar{m},T} \right]^{\frac{1}{\gamma-1}} & \text{for } m = \bar{m} \end{cases}.$$

3. Given the policy functions at $T - dt$ and the interest rate guess, solve for the

value functions $v_{m,T-dt}$:

$$\begin{aligned}
v_{m,T-dt} = & \left(\pi(m) - \alpha \frac{x_{m,T-dt}^\gamma}{\gamma} \right) dt \\
& + e^{-r_T dt} (x_{m,T-dt} dt \sum_{n=m+1}^{\bar{m}} \mathbb{F}_{m,T-dt}(n) [v_{n,T}(\lambda^{n-m})^{\frac{1}{\beta}-1} - v_{m,T}] \\
& + x_{-m,T-dt} dt \sum_{-m+1}^{\bar{m}} \mathbb{F}_{-m,T-dt}(n) [v_{-n,T} - v_{m,T}] \\
& + \delta_e dt (0 - v_{m,T}) + v_{m,T}).
\end{aligned}$$

4. Repeat this backwards iteration for $x_{m,t}$ and $v_{m,t}$ until $t = 1$.
5. Initialize a panel of firms with stationary distribution μ_{mt} consistent with the initial steady state. Given $\{x_{m,t}\}_{t=1}^{T-dt}$, simulate forward to obtain aggregate R_t , K_t , Y_t and obtain household consumption from the resource constraint: $C_t = Y_t - K_t - R_t$ for $t \in \{t, \dots, T\}$. Repeat this forward simulation S times.
6. Check if the conjectured interest rate sequence $\mathbf{r}$ is consistent with the resulting sequence of consumption growth rates using the households' Euler equation.
7. Update the guess of $\mathbf{r}$ to $\mathbf{r}_{new}$ using the average of the implied sequence of interest rates from the Euler equation over the S simulations.
8. Repeat until $|\mathbf{r}_{new} - \mathbf{r}| < \epsilon_{tol}$ for some small tolerance value ϵ_{tol} .

At each iteration a panel of 3,000 firms is simulated 50 times over 300 years with $dt = 0.1$.

<u>Moment</u>	<u>Model</u>		<u>Data</u>		<u>Effect of each parameter</u>						
	1990s	2000s	1990s	2000s	ρ	ϵ	η	δ_e	λ	α	ϕ
TFP growth	1.58	0.66	1.57	0.66	1.45	1.56	2.20	1.64	1.89	1.40	0.38
Leader market share	42.62	48.12	43.23	48.12	42.57	43.37	42.70	42.64	43.70	42.59	51.22
R&D/GDP	1.76	1.92	1.80	1.89	1.49	1.74	4.54	1.91	2.09	1.61	0.62
Profits/GDP	7.11	7.43	7.71	9.20	7.37	7.06	4.35	6.97	6.98	7.25	9.11
Leadership turnover	14.16	9.96	13.74	9.28	13.84	14.11	15.65	13.42	14.51	13.69	10.04
Pat. qual., followers	16.67	9.49	19.02	11.86	17.68	17.15	15.04	16.19	18.05	17.21	6.88
Pat. qual., leaders	8.44	7.43	6.87	4.91	8.66	8.64	8.05	8.33	9.24	8.53	6.42
Patent qual., std. dev.	27.61	19.81	25.75	19.49	29.17	28.52	25.22	26.86	30.58	28.47	12.83

Table C.1: Effect of each estimated parameter change in Table 5 on the model steady state, holding other parameters fixed at estimated 1990s values. Decompositions follow the formula in equation 15.

C.4 Decomposition Table